

A TEMPERED DROP LEMMA FOR BAYESIAN VARIABLE SELECTION

BY YVANN LE FAY^{1,a}

¹CREST, ENSAE, IP Paris, ^ayvann.lefay@ensae.fr

Sequential Monte Carlo samplers for the Bayesian variable selection model of Yang, Wainwright and Jordan (2016) travel along the tempering path obtained by raising the marginal likelihood to a power $\lambda \in [0, 1]$, and deliver, besides the posterior, the normalising constant that a Markov chain alone does not give. Their cost is governed by the mixing of the Metropolis–Hastings kernel at every intermediate temperature, whereas Yang, Wainwright and Jordan (2016) bound the spectral gap at $\lambda = 1$ only, by a canonical path ensemble directed at the true model γ^* . We ask what survives at $\lambda < 1$, where the sparsity prior is untempered while the likelihood is damped, so that the mode of the target slides from the null model at $\lambda = 0$ to γ^* at $\lambda = 1$.

Three things. Above an explicit critical temperature λ_* their argument transposes with no change beyond the substitution of the tempered posterior for the posterior: the same drop bound, a tempered concentration $\pi_n^{(\lambda)}(\gamma^* | Y) \geq 1/4$, a spectral gap at least $1/(64ps_0^2)$ and a mixing time $\mathcal{O}(ps_0^2\{\alpha(n + s_0)\log p + \log(1/\epsilon)\})$, with constants free of λ ; at $\lambda = 1$ these are their Theorems 1 and 2. The restriction to $\lambda \geq \lambda_*$ is a property of the problem and not of the proof: an exact identity shows that no ensemble directed at γ^* can be non-increasing at every temperature. Moving the centre instead, through a map whose fixed point is defined by a pair of temperature-dependent thresholds, gives a gap at least $1/\{40ps_0(2s^* + 1)\}$ at every temperature outside a computable union of at most s^* short windows. Inside a window the covariates it leaves undecided are nearly neutral, and an ensemble that declines to commit to a value for them is charged only for their interaction with each other, not for their number. The last two cost one condition on the design that Yang, Wainwright and Jordan (2016) do not make.

1. Introduction. Yang, Wainwright and Jordan (2016) — YWJ throughout — study high-dimensional Bayesian variable selection with a truncated sparsity prior, and prove two things about it. The posterior concentrates on the true model, and a particular Metropolis–Hastings chain targeting that posterior mixes in a number of steps linear in the number p of covariates, up to logarithmic factors. The second statement is the delicate one, and they show by example that it does not follow from the first: a posterior can be sharply concentrated and still be, for a local chain, effectively multimodal. Their proof of it is a canonical path argument, and the paths are the trajectories of a greedy variable-selection algorithm, all of them directed at the true model γ^* .

This note asks what becomes of that argument when the posterior is tempered: when the marginal likelihood is raised to a power $\lambda \in [0, 1]$ and the sparsity prior is left alone, so that the target interpolates between the prior at $\lambda = 0$ and the posterior at $\lambda = 1$.

Why the tempered targets, and not only the posterior. Two quantities are wanted from this model, and only one of them is delivered by a Markov chain. The first is the posterior over models, which an MCMC run approximates directly. The second is the normalising constant

$$Z = \sum_{\gamma} \pi(\gamma) \mathcal{L}_n(Y | \gamma),$$

the sum running over the models the prior retains — the marginal likelihood of the data under the whole hierarchical model — which a single MCMC run does not give at all: the chain sees ratios of posterior masses, and Z cancels from every one of them.

Z is a function of the hyperparameters (g, κ) of (4), and the theory does not choose them: Assumption 3.3 constrains κ and the exponent α of $g \asymp p^{2\alpha}$ only through $\kappa + \alpha \geq 8(L + \tilde{L}) + 2$, and $L, \tilde{L}$ are unknown. They must therefore be chosen from the data, by maximising $Z(g, \kappa)$ or averaging against a hyperprior — a marginal-likelihood computation over a grid.

2. Setting. We keep the model of YWJ and their transition function, and change one thing only: the marginal likelihood is raised to the power λ . Every other quantity that occurs is written out rather than named.

2.1. The model. Let $Y \in \mathbb{R}^n$ be a response vector and $X \in \mathbb{R}^{n \times p}$ a design matrix whose columns $X_1, \dots, X_p$ are normalised so that $\|X_j\|_2^2 = n$. The data are generated by

$$(1) \quad Y = X\beta^* + w, \quad w \sim \mathcal{N}(0, \sigma_0^2 I_n),$$

with $\beta^* \in \mathbb{R}^p$ and $\sigma_0 > 0$ unknown.

A subset of covariates is encoded by a binary vector $\gamma \in \{0, 1\}^p$, and we adopt the dualistic view of YWJ, using the same letter for the vector and for the subset $\{j \in [p] : \gamma_j = 1\}$ of $[p] := \{1, \dots, p\}$ that it indicates, so that $j \in \gamma$, $\gamma \cup \{j\}$, $\gamma \subseteq \gamma'$ and $|\gamma| = \sum_j \gamma_j$ all have their obvious meaning. We write X_γ for the submatrix of X with columns indexed by γ and

$$(2) \quad \Phi_\gamma := X_\gamma(X_\gamma^\top X_\gamma)^{-1} X_\gamma^\top, \quad \Phi_\emptyset := 0,$$

for the orthogonal projection onto their span. We use repeatedly, and without further comment, the fact that if $\gamma \subseteq \gamma'$ then $\Phi_{\gamma'} \Phi_\gamma = \Phi_\gamma$, so that $\Phi_{\gamma'} - \Phi_\gamma$ is itself an orthogonal projection, and that $\|\Phi_\gamma v\|_2 \leq \|v\|_2$ for every v .

Fix a threshold $C_\beta > 0$. The set of *influential* covariates is

$$(3) \quad S := \{j \in [p] : |\beta_j^*| \geq C_\beta\}, \quad S^c := [p] \setminus S, \quad s^* := |S|,$$

and γ^* denotes the binary vector indicating S , so that $X_S \beta_S^* = X_{\gamma^*} \beta_{\gamma^*}^*$; we use the two notations interchangeably, as YWJ do. The coefficients $\beta_{S^c}^*$ off the support need not vanish; they are only required to be small, which is Assumption 3.1 below.

The Bayesian model puts a flat prior $\pi(\phi) \propto 1/\phi$ on the noise precision, Zellner's g -prior $\beta_\gamma | \gamma, \phi \sim \mathcal{N}(0, g\phi^{-1}(X_\gamma^\top X_\gamma)^{-1})$ on the regression coefficients, and the truncated sparsity prior

$$(4) \quad \pi(\gamma) \propto p^{-\kappa|\gamma|} \mathbb{I}[|\gamma| \leq s_0],$$

on the model space $\mathcal{M}(s_0) := \{\gamma \in \{0, 1\}^p : |\gamma| \leq s_0\}$, where $\kappa > 0$ and $s^* < s_0 < n$. Integrating out (β_γ, ϕ) gives the marginal likelihood of their equation (13),

$$(5) \quad \mathcal{L}_n(Y | \gamma) \propto \frac{1}{\|Y\|_2^n} \cdot \frac{(1+g)^{-|\gamma|/2}}{\{1+g(1-R_\gamma^2)\}^{n/2}}, \quad R_\gamma^2 := \frac{Y^\top \Phi_\gamma Y}{\|Y\|_2^2}.$$

We assume $Y \neq 0$, an event of probability one, so that R_γ^2 is well defined.

2.2. The tempered posterior. For $\lambda \geq 0$ define

$$(6) \quad \pi_n^{(\lambda)}(\gamma | Y) := \frac{\pi(\gamma) \mathcal{L}_n(Y | \gamma)^\lambda}{\sum_{\bar{\gamma} \in \mathcal{M}(s_0)} \pi(\bar{\gamma}) \mathcal{L}_n(Y | \bar{\gamma})^\lambda}, \quad \gamma \in \mathcal{M}(s_0).$$

Since $\mathcal{M}(s_0)$ is finite and $\mathcal{L}_n(Y | \cdot) > 0$ on it, the denominator lies in $(0, \infty)$ and $\pi_n^{(\lambda)}(\cdot | Y)$ is a well-defined probability measure with full support on $\mathcal{M}(s_0)$, for every real λ . At $\lambda = 0$ it is the sparsity prior π ; at $\lambda = 1$ it is the posterior $\pi_n(\cdot | Y)$ of YWJ.

For every $\lambda \geq 0$ and every $\gamma, \gamma' \in \mathcal{M}(s_0)$,

$$(7) \quad \log \frac{\pi_n^{(\lambda)}(\gamma | Y)}{\pi_n^{(\lambda)}(\gamma' | Y)} = (|\gamma'| - |\gamma|) \left\{ \kappa \log p + \frac{\lambda}{2} \log(1 + g) \right\} + \frac{\lambda n}{2} \log \frac{g^{-1} + 1 - R_{\gamma'}^2}{g^{-1} + 1 - R_{\gamma}^2}.$$

The first term of (7) is a *penalty*, affine in λ with intercept $\kappa \log p$ because the sparsity prior is untempered while the Occam factor $(1 + g)^{-1/2}$ is; the second is a *fit* term, proportional to λ and switched off at $\lambda = 0$. Every statement below compares the two, and the whole difference with the untempered argument of YWJ is that at $\lambda < 1$ the fit term is damped while $\kappa \log p$ is not. For $|\gamma'| - |\gamma| = 1$, (7) reads

$$(8) \quad \log \frac{\pi_n^{(\lambda)}(\gamma \cup \{j\} | Y)}{\pi_n^{(\lambda)}(\gamma | Y)} = \lambda \log \frac{\mathcal{L}_n(Y | \gamma \cup \{j\})}{\mathcal{L}_n(Y | \gamma)} - \kappa \log p.$$

3. Known results. We restate the assumptions of YWJ. Two of them must be imposed slightly above the truncation level s_0 , because Lemma A.1 below compares a model $\gamma \in \mathcal{M}(s_0)$ with $\gamma \cup \gamma^*$, whose size may exceed s_0 . We therefore impose their Assumptions B and D at the level

$$(9) \quad s := s_0 + s^* \leq 2s_0,$$

and work with the model class $\mathcal{M}(s)$. Since $s^* < s_0$ this strengthens their hypotheses by a factor at most two in the model size, and by nothing else.

ASSUMPTION 3.1 (Size of β^* ; YWJ, Assumption A). There is a constant $\tilde{L} < \infty$ with

$$\|X\beta^*\|_2^2 \leq g\sigma_0^2 \log p, \quad \|X_{S^c}\beta_{S^c}^*\|_2^2 \leq \tilde{L}\sigma_0^2 \log p.$$

ASSUMPTION 3.2 (Design; YWJ, Assumption B at level s). There are constants $\nu > 0$ and $L < \infty$ such that, with $Z \sim \mathcal{N}(0, I_n)$,

$$\min_{|\gamma| \leq s} \lambda_{\min}(n^{-1}X_{\gamma}^{\top}X_{\gamma}) \geq \nu, \quad \text{and} \\ \mathbb{E}_Z \left[\max_{|\gamma| \leq s} \max_{k \in [p] \setminus \gamma} n^{-1/2} |\langle (I - \Phi_{\gamma})X_k, Z \rangle| \right] \leq \frac{1}{2} \sqrt{L\nu \log p}.$$

ASSUMPTION 3.3 (Hyperparameters; YWJ, Assumption C). There is $\alpha \geq 1/2$ with $g \asymp p^{2\alpha}$ and $g \geq \max\{n, p^{2\alpha}\}$, and $\kappa + \alpha \geq 8(L + \tilde{L}) + 2$.

ASSUMPTION 3.4 (Sparsity control; YWJ, Assumption D at level s). With $\omega(X) := \max_{|\gamma| \leq s} \|(X_{\gamma}^{\top}X_{\gamma})^{-1}X_{\gamma}^{\top}X_{S \setminus \gamma}\|_{\text{op}}^2$,

$$(10) \quad \max\{1, (4\nu^{-2}\omega(X) + 1)s^*\} \leq s_0 \leq s \leq \frac{1}{256} \left\{ \frac{n}{\log p} - 32\tilde{L} \right\}.$$

The only difference with YWJ here is the constant 4 in place of their 2 on the left, which buys the margin that Lemma A.4b needs and that their proof of the saturated case does not in fact have (see Remark A.1); on the right they have $\frac{1}{32}\{n/\log p - 8\tilde{L}\}$, at the level s_0 .

ASSUMPTION 3.5 ($\beta_{\min}$ condition; YWJ, equation (23)). $n\nu^2 C_{\beta}^2 \geq 128(L + \tilde{L})\sigma_0^2 \log p$.

This is their $\beta_{\min}$ condition $C_{\beta}^2 \geq c_0(L + \tilde{L} + \alpha + \kappa)\sigma_0^2 \log p/n$ with the factor ν^2 folded in, restricted to the part of it that controls the noise; the part involving $\alpha + \kappa$ reappears, quantified, in Corollary 4.1.

LEMMA 3.1 (Good event; YWJ, Lemma 3). *Define*

$$(11) \quad \begin{aligned} \mathcal{A}_n &:= \left\{ \max_{\gamma_2 \subset \gamma_1 \in \mathcal{M}(s)} \frac{w^\top (\Phi_{\gamma_1} - \Phi_{\gamma_2}) w}{|\gamma_1| - |\gamma_2|} \leq L \sigma_0^2 \log p \right\}, \quad \mathcal{B}_n := \left\{ \max_{\gamma \in \mathcal{M}(s)} \frac{w^\top \Phi_\gamma w}{|\gamma|} \leq 8 \sigma_0^2 \log p \right\}, \\ \mathcal{C}_n &:= \left\{ \left| \frac{\|w\|_2^2}{n \sigma_0^2} - 1 \right| \leq \frac{1}{2} \right\}, \quad \mathcal{D}_n := \left\{ \frac{\|Y\|_2^2}{g} \leq (2 \log p + 3) \sigma_0^2 \right\}. \end{aligned}$$

Under Assumptions 3.1 to 3.4, $\mathbb{P}(\mathcal{A}_n \cap \mathcal{B}_n \cap \mathcal{C}_n \cap \mathcal{D}_n) \geq 1 - 6p^{-c}$ for a universal constant $c > 0$.

This is Lemma 3 of YWJ with their model class $\mathcal{M}(s_0)$ replaced by $\mathcal{M}(s)$, which their proof allows because the truncation level enters it only through that class.

From here on, every statement made “on the good event” is a deterministic statement about a fixed realisation of (Y, w) lying in $\mathcal{A}_n \cap \mathcal{B}_n \cap \mathcal{C}_n \cap \mathcal{D}_n$. No further probabilistic argument occurs.

Under Assumptions 3.1 to 3.5, YWJ prove two separate statements about the posterior $\pi_n(\cdot | Y) = \pi_n^{(1)}(\cdot | Y)$: that it concentrates on γ^* , and that the Metropolis–Hastings chain targeting it mixes in $\tilde{O}(ps_0^2)$ steps.

3.1. Posterior concentration.

THEOREM 3.1 (Posterior concentration; YWJ, Theorem 1). *Suppose Assumption 3.1 and Assumption 3.3; Assumption 3.2 at the level $s = 2(\kappa + \alpha + \tilde{L} + 1) \max\{1, s^*\}$; the sparsity prior (4) left untruncated, $s_0 = p$, together with*

$$\begin{aligned} \max\{1, s^*\} &\leq \frac{1}{32} \left\{ \frac{n}{\log p} - 8\tilde{L} \right\} \quad \text{in place of Assumption 3.4,} \\ \text{and} \quad C_\beta^2 &\geq \frac{128}{\nu^2} (L + \tilde{L} + \alpha + \kappa) \frac{\sigma_0^2 \log p}{n}. \end{aligned}$$

Then $\pi_n(\gamma^* | Y) \geq 1 - c_1 p^{-1}$ with probability at least $1 - c_2 p^{-c_3}$, the c_i being universal.

3.2. *The Metropolis–Hastings chain.* The sampler is that of YWJ, and it depends on the distribution it targets only through the acceptance ratio. We therefore set it up once, for an arbitrary probability measure μ with full support on $\mathcal{M}(s_0)$, and carry no temperature until there is something for it to do. For $\gamma \in \mathcal{M}(s_0)$ put

$$(12) \quad \begin{aligned} \mathcal{N}_1(\gamma) &:= \{\gamma' : d_H(\gamma, \gamma') = 1\}, \\ \mathcal{N}_2(\gamma) &:= \{\gamma' : d_H(\gamma, \gamma') = 2 \text{ and } \exists (k, \ell) \in \gamma \times ([p] \setminus \gamma) \text{ with } \gamma' = \gamma \cup \{\ell\} \setminus \{k\}\}, \end{aligned}$$

and $\mathcal{N}(\gamma) := \mathcal{N}_1(\gamma) \cup \mathcal{N}_2(\gamma)$. The chain proposes, with probability 1/2 each, a single flip of a uniform coordinate of $[p]$ or an exchange of a uniform pair in $\gamma \times ([p] \setminus \gamma)$, and accepts by the Metropolis rule; a proposal outside $\mathcal{M}(s_0)$ is rejected, which is automatic since μ vanishes there. Writing $\tilde{\mathbf{P}}$ for the resulting transition matrix,

$$(13) \quad \tilde{\mathbf{P}}(\gamma, \gamma') = \begin{cases} \frac{1}{2p} \min\left\{1, \frac{\mu(\gamma')}{\mu(\gamma)}\right\}, & \gamma' \in \mathcal{N}_1(\gamma) \cap \mathcal{M}(s_0), \\ \frac{1}{2|\gamma|(p - |\gamma|)} \min\left\{1, \frac{\mu(\gamma')}{\mu(\gamma)}\right\}, & \gamma' \in \mathcal{N}_2(\gamma), \\ 0, & \gamma' \notin \mathcal{N}(\gamma) \cup \{\gamma\}, \end{cases}$$

the diagonal being fixed by $\sum_{\gamma'} \tilde{\mathbf{P}}(\gamma, \gamma') = 1$, and we analyse the lazy chain

$$(14) \quad \mathbf{P} := \frac{1}{2}\mathbf{I} + \frac{1}{2}\tilde{\mathbf{P}}, \quad \mathbf{Q}(\gamma, \gamma') := \mu(\gamma) \mathbf{P}(\gamma, \gamma').$$

LEMMA 3.2 (The kernel). *$\mathbf{P}$ is μ -reversible, irreducible and aperiodic on $\mathcal{M}(s_0)$, and non-negative definite, so that its smallest eigenvalue is at least 0 and $\text{Gap}(\mathbf{P}) = 1 - \lambda_2(\mathbf{P})$ is its absolute spectral gap. Moreover $\mathbf{Q}$ is symmetric and, for every $\gamma \in \mathcal{M}(s_0)$ and every $\gamma' \in \mathcal{N}(\gamma) \cap \mathcal{M}(s_0)$,*

$$(15) \quad \mathbf{Q}(\gamma, \gamma') \geq \frac{1}{4ps_0} \min\{\mu(\gamma), \mu(\gamma')\}.$$

Convention. From Section 4 onwards the target is $\mu = \pi_n^{(\lambda)}(\cdot | Y)$, and we then write $\tilde{\mathbf{P}}_\lambda$, $\mathbf{P}_\lambda$, $\mathbf{Q}_\lambda$ and $\tau_\epsilon(\mathbf{P}_\lambda)$ for the objects just defined, to record which target they refer to; at $\lambda = 1$ they are the chain of YWJ. Only the acceptance ratio in (13) changes with the temperature. The proposal, the laziness, the underlying graph and every statement of the present subsection and of Section 3.3 do not, and are used below without a subscript wherever the target is immaterial.

3.3. *Canonical paths and Sinclair's bound.* The spectral gap of a reversible chain can be bounded below by exhibiting, for each ordered pair of states, a path between them, and checking that no edge of the graph is asked to carry too much probability mass.

Identify $\mathbf{P}$ with the weighted directed graph on $\mathcal{M}(s_0)$ whose edges are the ordered pairs $e = (\gamma, \gamma')$ with $\mathbf{P}(\gamma, \gamma') > 0$, each carrying the weight $\mathbf{Q}(e)$ of (14), which is symmetric by Lemma 3.2. A *canonical path ensemble* $\mathcal{T}$ assigns to every ordered pair (γ, γ') of distinct states a simple path $T_{\gamma, \gamma'}$ in that graph. Its two characteristics are the *path congestion* and the *maximal length*,

$$(16) \quad \rho(\mathcal{T}) := \max_e \frac{1}{\mathbf{Q}(e)} \sum_{T_{\gamma, \gamma'} \ni e} \mu(\gamma) \mu(\gamma'), \quad \ell(\mathcal{T}) := \max_{\gamma \neq \gamma'} |T_{\gamma, \gamma'}|,$$

the maximum being over directed edges e , and Sinclair's bound (Sinclair, 1992, Corollary 6') reads

$$(17) \quad \text{Gap}(\mathbf{P}) \geq \frac{1}{\rho(\mathcal{T}) \ell(\mathcal{T})}.$$

An arbitrary ensemble is unmanageable, and the device that makes (17) usable is to generate the ensemble from a map. Following YWJ, call $\mathcal{G} : \mathcal{M}(s_0) \rightarrow \mathcal{M}(s_0)$ a *valid transition function* with centre γ^* if $\mathcal{G}(\gamma^*) = \gamma^*$ and, for every $\gamma \neq \gamma^*$, the pair $(\gamma, \mathcal{G}(\gamma))$ is an edge of the graph and $d_H(\mathcal{G}(\gamma), \gamma^*) < d_H(\gamma, \gamma^*)$. The orbit $\gamma \rightarrow \mathcal{G}(\gamma) \rightarrow \mathcal{G}^2(\gamma) \rightarrow \dots$ then reaches γ^* in finitely many steps and is taken as T_{γ, γ^*} ; for a general pair, $T_{\gamma, \gamma'}$ follows T_{γ, γ^*} up to the first state it shares with T_{γ', γ^*} and then follows T_{γ', γ^*} backwards. This is Lemma 1 of YWJ. The ensemble so obtained is *memoryless*: how a path continues depends on the state it has reached and on nothing else.

DEFINITION 3.1 (Precedents; YWJ, Section 4.1). For $\gamma \in \mathcal{M}(s_0)$ put

$$(18) \quad \Lambda(\gamma) := \{\bar{\gamma} \in \mathcal{M}(s_0) : \mathcal{G}^k(\bar{\gamma}) = \gamma \text{ for some } k \geq 0\},$$

the states whose canonical path to γ^* passes through γ ; it contains γ itself, at $k = 0$. For $\bar{\gamma} \in \Lambda(\gamma)$ we write $k(\bar{\gamma})$ for the smallest such k , and $\Lambda_k(\gamma) := \{\bar{\gamma} : k(\bar{\gamma}) = k\}$.

Memorylessness now does all the combinatorial work. A directed edge $e = (\bar{\gamma}, \mathcal{G}(\bar{\gamma}))$ is traversed in the direction of $\mathcal{G}$, so it can occur only in the first segment of a path, which forces the source of that path to be a precedent of $\bar{\gamma}$; this is Lemma A.7c. Since $\sum_{\gamma'} \mu(\gamma') = 1$, the numerator of (16) is at most $\mu[\Lambda(\bar{\gamma})]$; the reverse orientation $(\mathcal{G}(\bar{\gamma}), \bar{\gamma})$ obeys the same bound, by the same argument with the roles of the two endpoints exchanged and because $\mathbf{Q}$ is symmetric. Bounding the denominator by (15),

$$(19) \quad \rho(\mathcal{T}) \leq 4ps_0 \max_{\bar{\gamma} \neq \gamma^*} \max \left\{ 1, \frac{\mu(\bar{\gamma})}{\mu(\mathcal{G}(\bar{\gamma}))} \right\} \cdot \frac{\mu[\Lambda(\bar{\gamma})]}{\mu(\bar{\gamma})}.$$

The length is combinatorial too: each step of $\mathcal{G}$ decreases $d_H(\cdot, \gamma^*)$, so $\ell(\mathcal{T}) \leq 4s_0$, by Lemma A.7b.

Nothing so far has mentioned the temperature, and nothing has used any property of μ . The whole dependence on the target sits in the two factors of (19), and it is there — and only there — that λ will enter. Written out for the target that concerns us, those two factors ask for the two halves of Lemma 4 of YWJ:

- (D $_\lambda$) a *drop bound*: the target does not increase along $\mathcal{G}$, that is $\pi_n^{(\lambda)}(\gamma | Y) \leq \pi_n^{(\lambda)}(\mathcal{G}(\gamma) | Y)$ for every $\gamma \neq \gamma^*$, and preferably by a definite factor $p^{-\eta}$;
- (P $_\lambda$) a *precedent mass bound*: $\pi_n^{(\lambda)}[\Lambda(\gamma) | Y] \leq c \pi_n^{(\lambda)}(\gamma | Y)$ for a constant c and every γ .

Given both, (19) gives $\rho(\mathcal{T}) \leq 4cps_0$ and (17) gives a gap of order $1/(ps_0^2)$. Moreover (P $_\lambda$) is a consequence of (D $_\lambda$) and of counting alone: a precedent at distance k carries at most $p^{-\eta k}$ times the mass of γ , by telescoping the drop bound along the path, and there are at most $\{(s^* + 1)p\}^k$ of them, by Lemma A.6; the geometric series converges as soon as η exceeds 1 by a little. So a single inequality, the drop bound, stands between the assumptions and the spectral gap.

3.4. The transition function. We now recall the map that drives the canonical path ensemble of YWJ. It is defined in terms of the design and of β^* only, and does *not* depend on λ .

DEFINITION 3.2 (States; YWJ, Section 4.1). A state $\gamma \in \mathcal{M}(s_0)$ with $\gamma \neq \gamma^*$ is

- *overfitted* if $\gamma \supseteq \gamma^*$, and *underfitted* otherwise, that is if $\gamma^* \setminus \gamma \neq \emptyset$;
- *saturated* if $|\gamma| = s_0$, and *unsaturated* if $|\gamma| < s_0$.

An overfitted $\gamma \neq \gamma^*$ satisfies $\gamma \setminus \gamma^* \neq \emptyset$, and an underfitted state can only exist if $s^* \geq 1$.

DEFINITION 3.3 (Greedy transition function; YWJ, Section 4.1). For $\gamma \in \mathcal{M}(s_0) \setminus \{\gamma^*\}$ define $\mathcal{G}(\gamma)$ by the first applicable rule:

(G1) γ *overfitted*. Let

$$(20) \quad \ell_\gamma \in \arg \min_{\ell \in \gamma \setminus \gamma^*} \left\{ \|\Phi_\gamma X_{\gamma^*} \beta_{\gamma^*}^*\|_2^2 - \|\Phi_{\gamma \setminus \{\ell\}} X_{\gamma^*} \beta_{\gamma^*}^*\|_2^2 \right\},$$

the least influential uninfluential covariate present, and set $\mathcal{G}(\gamma) := \gamma \setminus \{\ell_\gamma\}$.

(G2) γ *underfitted and unsaturated*. Let

$$(21) \quad j_\gamma \in \arg \max_{j \in \gamma^* \setminus \gamma} \|\Phi_{\gamma \cup \{j\}} X_{\gamma^*} \beta_{\gamma^*}^*\|_2^2,$$

the missing influential covariate explaining most signal, and set $\mathcal{G}(\gamma) := \gamma \cup \{j_\gamma\}$.

(G3) γ *underfitted and saturated*. With j_γ as in (21) and $\tilde{\gamma} := \gamma \cup \{j_\gamma\}$, let

$$(22) \quad k_\gamma \in \arg \min_{k \in \gamma \setminus \gamma^*} \left\{ \|\Phi_{\tilde{\gamma}} X_{\gamma^*} \beta_{\gamma^*}^*\|_2^2 - \|\Phi_{\tilde{\gamma} \setminus \{k\}} X_{\gamma^*} \beta_{\gamma^*}^*\|_2^2 \right\},$$

and set $\mathcal{G}(\gamma) := \tilde{\gamma} \setminus \{k_\gamma\} = \gamma \cup \{j_\gamma\} \setminus \{k_\gamma\}$.

Ties in (20)–(22) are broken by the lexicographic order, and $\mathcal{G}(\gamma^*) := \gamma^*$.

Rules (G1) and (G2) decrease $d_H(\cdot, \gamma^*)$ by one and rule (G3) decreases it by two, so $\mathcal{G}$ is a valid transition function in the sense of YWJ’s Lemma 1 and generates a memoryless canonical path ensemble directed at γ^* , of maximal path length at most $2s_0$.

We record the three size changes once and for all, since they are what (7) feeds on:

$$(23) \quad |\mathcal{G}(\gamma)| - |\gamma| = \begin{cases} -1 & \text{under rule (G1),} \\ +1 & \text{under rule (G2),} \\ 0 & \text{under rule (G3).} \end{cases}$$

3.5. From the spectral gap to the mixing time. A spectral gap is not yet a mixing time. For the lazy chain $\mathbf{P}$ of Section 3.2, or indeed any lazy reversible chain with stationary law μ on a finite state space, the ϵ -mixing time $\tau_\epsilon := \max_\gamma \min\{t : \|\mathbf{P}^{t'}(\gamma, \cdot) - \mu\|_{\text{TV}} \leq \epsilon \forall t' \geq t\}$ obeys

$$(24) \quad \tau_\epsilon \leq \frac{\log[1/\min_\gamma \mu(\gamma)] + \log(1/\epsilon)}{\text{Gap}(\mathbf{P})},$$

which is the upper half of the sandwich used by YWJ in their equation (14); see Sinclair (1992).

The numerator of (24) is where posterior concentration re-enters, and it enters modestly. YWJ bound $\min_\gamma \pi_n(\gamma | Y)$ below by factorising $\pi_n(\gamma | Y) = \pi_n(\gamma^* | Y) \cdot \{\pi_n(\gamma | Y)/\pi_n(\gamma^* | Y)\}$: the first factor is bounded below by Theorem 3.1 and the second by the explicit ratio (7) at $\lambda = 1$, whose worst case costs a factor $p^{-\kappa s_0} (1+g)^{-(n+s_0)/2}$ — this is Lemma 5.2 at $\lambda = 1$. Taking logarithms, the numerator is $\mathcal{O}\{(\alpha n + \alpha s_0 + \kappa s_0) \log p + \log(1/\epsilon)\}$, and their gap bound $\text{Gap}(\mathbf{P}) \geq 1/(24ps_0^2)$ turns this into their Theorem 2,

$$\tau_\epsilon \leq 12ps_0^2 \{(\alpha n + \alpha s_0 + 2\kappa s_0) \log p + \log(1/\epsilon) + \log 2\}.$$

Two features of this last step are worth isolating, because both survive tempering unchanged. The concentration statement is used only through the smallest mass, so any constant lower bound on $\pi_n(\gamma^* | Y)$ would serve equally well: the $(n + s_0) \log p$ term comes from the ratio, not from the quality of the concentration. And that term is a burn-in cost; started at γ^* rather than at the worst state, the chain does not pay it.

4. The drop bound at temperature λ . We come to (D_λ) , for $\pi_n^{(\lambda)}(\cdot | Y)$ along the map $\mathcal{G}$ of Definition 3.3. This section establishes it, together with its consequence (P_λ) , above an explicit threshold, and shows that no such threshold can be dispensed with.

THEOREM 4.1 (Drop bound at temperature λ). *Assume Assumptions 3.1 to 3.5, and work on the good event. Let $\lambda \geq 0$ and let $\gamma \in \mathcal{M}(s_0)$ with $\gamma \neq \gamma^*$. Then, if γ is overfitted,*

$$(25) \quad \log \frac{\pi_n^{(\lambda)}(\gamma | Y)}{\pi_n^{(\lambda)}(\mathcal{G}(\gamma) | Y)} \leq -\kappa \log p - \lambda \left\{ \frac{1}{2} \log(1+g) - 8(L + \tilde{L}) \log p \right\};$$

if γ is underfitted and unsaturated,

$$(26) \quad \log \frac{\pi_n^{(\lambda)}(\gamma | Y)}{\pi_n^{(\lambda)}(\mathcal{G}(\gamma) | Y)} \leq \kappa \log p + \lambda \left\{ \frac{1}{2} \log(1 + g) - \min \left\{ \frac{n\nu}{16s^*}, \frac{n\nu^2 C_\beta^2}{64\sigma_0^2} \right\} \right\};$$

and if γ is underfitted and saturated,

$$(27) \quad \log \frac{\pi_n^{(\lambda)}(\gamma | Y)}{\pi_n^{(\lambda)}(\mathcal{G}(\gamma) | Y)} \leq -\frac{\lambda}{2} \min \left\{ \frac{n\nu}{16s^*}, \frac{n\nu^2 C_\beta^2}{64\sigma_0^2} \right\}.$$

The proof is deferred to Sections A.4 to A.6. It is three applications of (7), one per rule of Definition 3.3, with the size change read off from (23); the estimates it uses are the deterministic bounds of Section A, none of which involves λ .

Read the three bounds as functions of λ . Each is affine, and the three intercepts and the three slopes tell the whole story.

- (i) (25) has intercept $-\kappa \log p$ and slope $-\{\frac{1}{2} \log(1 + g) - 8(L + \tilde{L}) \log p\}$, which is at most $-\{\alpha - 8(L + \tilde{L})\} \log p$ by Assumption 3.3. It uses no condition on λ whatsoever, and when $\alpha > 8(L + \tilde{L})$ its worst value on $[0, 1]$ is at $\lambda = 0$, where it reads $-\kappa \log p$. Deleting an uninfluential covariate is favourable at every temperature, and least favourable at the prior, where the sparsity exponent κ alone does the work.
- (ii) (26) has intercept $+\kappa \log p$ and slope $\frac{1}{2} \log(1 + g) - \min \left\{ \frac{n\nu}{16s^*}, \frac{n\nu^2 C_\beta^2}{64\sigma_0^2} \right\}$. The intercept is positive: at $\lambda = 0$ the bound is useless, and Proposition 4.1 shows that this is not a defect of the bound. The slope is negative precisely when the greedy gain exceeds the Occam factor of the g -prior, and then the bound becomes useful above the temperature at which the two cross.
- (iii) (27) has no intercept at all: the inclusion penalty cancels identically in an exchange, both models having the same size and the prior depending on the model only through its size, so that $\pi_n^{(0)}(\gamma | Y) = \pi_n^{(0)}(\gamma' | Y)$ exactly. Exchange is neutral under the prior and favourable as soon as the likelihood is switched on; the temperature it needs to reach a given exponent η is smaller than the one case (ii) needs by roughly a factor $(\kappa + \eta)/2\eta$.

Behind the three cases is the tension already visible in (7): the untempered prior charges $\kappa \log p$ per covariate at every temperature, while the fit term is damped by λ . The map $\mathcal{G}$ knows nothing of it, being built from X and β^* alone and directed at γ^* whatever the temperature. At $\lambda = 1$ that is the right destination; at $\lambda = 0$ the target is the prior, whose mode is the null model, and each of the s^* additions $\mathcal{G}$ must make divides the target by exactly p^κ . So no such map can satisfy (D_λ) at every temperature. What is true, and is what this section proves, is that the failure occupies an initial segment and nothing more: there is an explicit λ_* above which the argument of Section 3 transposes with no change beyond the substitution of $\pi_n^{(\lambda)}$ for π_n .

4.1. The critical temperature. We ask for the conclusion in the form “each ratio is at most $p^{-\eta}$ ” rather than merely “at most one”, because Section 3.3 needs a definite factor to sum the precedents against.

COROLLARY 4.1 (Critical temperature). *Under the hypotheses of Theorem 4.1, let $\eta > 0$ and suppose $\min \left\{ \frac{n\nu}{16s^*}, \frac{n\nu^2 C_\beta^2}{64\sigma_0^2} \right\} > \frac{1}{2} \log(1 + g)$. Set*

$$(28) \quad \lambda_*(\eta) := \max \left\{ \frac{(\kappa + \eta) \log p}{\min \left\{ \frac{n\nu}{16s^*}, \frac{n\nu^2 C_\beta^2}{64\sigma_0^2} \right\} - \frac{1}{2} \log(1 + g)}, \frac{2\eta \log p}{\min \left\{ \frac{n\nu}{16s^*}, \frac{n\nu^2 C_\beta^2}{64\sigma_0^2} \right\}} \right\}.$$

Then, for every $\lambda \geq \lambda_*(\eta)$ and every underfitted $\gamma \in \mathcal{M}(s_0)$,

$$(29) \quad \frac{\pi_n^{(\lambda)}(\gamma | Y)}{\pi_n^{(\lambda)}(\mathcal{G}(\gamma) | Y)} \leq p^{-\eta};$$

and the same bound holds for every overfitted $\gamma \in \mathcal{M}(s_0) \setminus \{\gamma^*\}$ and every $\lambda \geq 0$ such that $\kappa \log p + \lambda \{\frac{1}{2} \log(1+g) - 8(L + \tilde{L}) \log p\} \geq \eta \log p$. The left-hand side of that last condition being affine in λ , it holds for all $\lambda \in [0, 1]$ as soon as it holds at the two endpoints, where, using $\frac{1}{2} \log(1+g) \geq \alpha \log p$ from Assumption 3.3, it is implied by

$$(30) \quad \underbrace{\kappa \geq \eta}_{\lambda=0} \quad \text{and} \quad \underbrace{\kappa + \alpha \geq \eta + 8(L + \tilde{L})}_{\lambda=1}.$$

4.2. *The mass of the precedents.* Bound (P_λ) transposes with the same η that Corollary 4.1 uses.

COROLLARY 4.2 (Mass of the precedents at temperature λ). *Let $\eta > 0$ and $\lambda \geq 0$ satisfy the two conditions of Corollary 4.1, so that $\pi_n^{(\lambda)}(\gamma | Y) \leq p^{-\eta} \pi_n^{(\lambda)}(\mathcal{G}(\gamma) | Y)$ for every $\gamma \in \mathcal{M}(s_0) \setminus \{\gamma^*\}$. Suppose in addition that*

$$(31) \quad (s^* + 1)p^{1-\eta} \leq \frac{1}{2}, \quad \text{equivalently} \quad \eta \geq 1 + \frac{\log\{2(s^* + 1)\}}{\log p}.$$

Then, for every $\gamma \in \mathcal{M}(s_0)$,

$$(32) \quad \frac{\pi_n^{(\lambda)}[\Lambda(\gamma) | Y]}{\pi_n^{(\lambda)}(\gamma | Y)} \leq \frac{1}{\{1 - (s^* + 1)p^{1-\eta}\} \{1 - p^{1-\eta}\}} \leq 4.$$

4.3. *The restriction on λ is real.* Item (ii) above asserted that the positive intercept of (26) belongs to the problem and not to the estimate. The following identity, valid for every design and every realisation of the data, is that assertion.

PROPOSITION 4.1 (Exact threshold). *Let $\gamma \in \mathcal{M}(s_0)$ be underfitted and unsaturated, so that $\mathcal{G}(\gamma) = \gamma \cup \{j_\gamma\}$. Then, for every $\lambda \geq 0$,*

$$(33) \quad \log \frac{\pi_n^{(\lambda)}(\gamma | Y)}{\pi_n^{(\lambda)}(\mathcal{G}(\gamma) | Y)} = \kappa \log p - \lambda \log \frac{\mathcal{L}_n(Y | \mathcal{G}(\gamma))}{\mathcal{L}_n(Y | \gamma)}.$$

Defining

$$(34) \quad \lambda_{\text{crit}} := \max \left\{ \frac{\kappa \log p}{\log \{ \mathcal{L}_n(Y | \mathcal{G}(\gamma)) / \mathcal{L}_n(Y | \gamma) \}} : \gamma \in \mathcal{M}(s_0) \text{ underfitted and unsaturated} \right\} \in (0, \infty],$$

the map $\mathcal{G}$ is non-decreasing for $\pi_n^{(\lambda)}(\cdot | Y)$ at every underfitted unsaturated state if and only if $\lambda \geq \lambda_{\text{crit}}$. Finally, on the good event and under the hypotheses of Theorem 4.1,

$$(35) \quad \lambda_{\text{crit}} \leq \frac{\kappa \log p}{\min \left\{ \frac{n\nu}{16s^*}, \frac{n\nu^2 C_\beta^2}{64\sigma_0^2} \right\} - \frac{1}{2} \log(1+g)} = \lambda_*(0)$$

whenever $\min \left\{ \frac{n\nu}{16s^*}, \frac{n\nu^2 C_\beta^2}{64\sigma_0^2} \right\} > \frac{1}{2} \log(1+g)$.

5. The spectral gap and the mixing time above λ_* . Corollaries 4.1 and 4.2 supply (D_λ) and (P_λ) above $\lambda_*(\eta)$, so the machinery of Section 3 runs unchanged. Nothing in this section is new relative to that one, and that is the point.

5.1. The ensemble and its congestion. Let $\mathcal{T}$ be the canonical path ensemble generated by $\mathcal{G}$ in the sense of Section 3.3: $\mathcal{G}$ is a valid transition function with centre γ^* , by the second remark after Definition 3.3, so T_{γ, γ^*} is the orbit $\gamma \rightarrow \mathcal{G}(\gamma) \rightarrow \dots \rightarrow \gamma^*$ and a general $T_{\gamma, \gamma'}$ is obtained from two such orbits. Every edge it uses joins some $\bar{\gamma}$ to $\mathcal{G}(\bar{\gamma})$, hence lies in $\mathcal{N}(\bar{\gamma})$ and carries positive weight by Lemma 3.2.

LEMMA 5.1 (Spectral gap at temperature λ). *Let $\eta > 0$ and $\lambda \geq 0$ satisfy the hypotheses of Corollary 4.2. Then, on the good event,*

$$(36) \quad \rho(\mathcal{T}) \leq 16 p s_0 \quad \text{and} \quad \text{Gap}(\mathbf{P}_\lambda) \geq \frac{1}{64 p s_0^2}.$$

Two features are worth recording. The bound does not involve η : the exponent only fixes the interval $[\lambda_*(\eta), 1]$ of temperatures on which the bound applies. And it does not involve λ either, so the same rate $1/(64 p s_0^2)$ holds simultaneously at every temperature in that interval — which is what an SMC complexity bound needs, since it requires a gap at each intermediate distribution and not only at the last.

5.2. Concentration at temperature λ . No statement of the form of Theorem 3.1 can hold at every $\lambda < 1$: at $\lambda = 0$ the target is the sparsity prior, which places mass of order $p^{-\kappa s^*}$ on γ^* . Above $\lambda_*(\eta)$, however, concentration comes for free from Corollary 4.2, because γ^* is the state all canonical paths end at.

PROPOSITION 5.1 (Concentration at temperature λ). *Under the hypotheses of Corollary 4.2, on the good event,*

$$(37) \quad \pi_n^{(\lambda)}(\gamma^* | Y) \geq \{1 - (s^* + 1)p^{1-\eta}\} \{1 - p^{1-\eta}\} \geq \frac{1}{4}.$$

The bound (37) is $1 - \mathcal{O}(s^* p^{1-\eta})$. Unlike Theorem 3.1 it is deterministic on the good event and holds at every $\lambda \geq \lambda_*(\eta)$, and it needs no separate argument: concentration and rapid mixing are here two readings of the same drop bound.

5.3. The smallest mass and the mixing time. Only one ingredient of the sandwich (24) is still missing, the smallest mass, and Proposition 5.1 supplies it: a target that puts at least a quarter of its mass on γ^* cannot put too little anywhere, because (7) bounds every ratio to γ^* explicitly.

LEMMA 5.2 (Lower bound on the smallest mass). *For every $\lambda \in [0, 1]$ and every $\gamma \in \mathcal{M}(s_0)$,*

$$(38) \quad \log \frac{\pi_n^{(\lambda)}(\gamma^* | Y)}{\pi_n^{(\lambda)}(\gamma | Y)} \leq \kappa s_0 \log p + \frac{\lambda(n + s_0)}{2} \log(1 + g).$$

Consequently, under the hypotheses of Corollary 4.2,

$$(39) \quad \min_{\gamma \in \mathcal{M}(s_0)} \pi_n^{(\lambda)}(\gamma | Y) \geq \frac{1}{4} p^{-\kappa s_0} (1 + g)^{-\lambda(n + s_0)/2}.$$

Feeding Lemmas 5.1 and 5.2 into (24) gives the tempered form of Theorem 2 of Yang, Wainwright and Jordan (2016).

THEOREM 5.1 (Mixing time at temperature λ). *Let $\eta > 0$ satisfy (31) and let $\lambda \in [0, 1]$ satisfy the two conditions of Corollary 4.1. Then, on the good event, for every $\epsilon \in (0, 1)$,*

$$(40) \quad \tau_\epsilon(\mathbf{P}_\lambda) \leq 64 p s_0^2 \left\{ \kappa s_0 \log p + \frac{\lambda(n + s_0)}{2} \log(1 + g) + \log \frac{1}{\epsilon} + \log 4 \right\}.$$

6. The temperature-adapted map. Proposition 4.1 says that no ensemble directed at γ^* can work below $\lambda_*(\eta)$: at $\lambda = 0$ the ratio along $\mathcal{G}$ is exactly p^κ . We therefore move the centre. The map $\mathcal{G}_\lambda$ of Definition 6.2 has fixed point c_λ^- in place of γ^* , and $c_0^- = \emptyset$ is the mode of $\pi_n^{(0)}(\cdot | Y) = \pi$.

Three of its four rules are uphill by the definition of $c_\lambda^\mp$, so only the undecided covariates $C_\lambda = c_\lambda^+ \setminus c_\lambda^-$ can cost anything. This section builds $\mathcal{G}_\lambda$ and bounds its drops and its precedents; Section 7 turns those into a spectral gap.

6.1. Log-gains. For $\gamma \in \mathcal{M}(s_0)$ with $|\gamma| < s_0$ and $j \in [p] \setminus \gamma$ put

$$(41) \quad \psi_j(\gamma) := \log \frac{\mathcal{L}_n(Y | \gamma \cup \{j\})}{\mathcal{L}_n(Y | \gamma)}, \quad \psi_j^- := \min_{v \subseteq \gamma^* \setminus \{j\}} \psi_j(v), \quad \psi_j^+ := \max_{v \subseteq \gamma^* \setminus \{j\}} \psi_j(v),$$

the last two for $j \in \gamma^*$ only; the extremes run over subsets of γ^* because rules (H2)–(H4) of Definition 6.2 fire only there. With this notation (7) with $|\gamma'| - |\gamma| = 1$ reads

$$(42) \quad \log \frac{\pi_n^{(\lambda)}(\gamma \cup \{j\} | Y)}{\pi_n^{(\lambda)}(\gamma | Y)} = \lambda \psi_j(\gamma) - \kappa \log p,$$

which is (8).

ASSUMPTION 6.1 (Null covariates are never worth buying). There is $a_0 < \infty$ such that, on the good event, $\psi_j(\gamma) \leq (a_0 - \alpha) \log p$ for every $\gamma \in \mathcal{M}(s_0)$ with $|\gamma| < s_0$ and every $j \in S^c \setminus \gamma$. We assume $\kappa \geq \eta$ and $\kappa + \alpha \geq \eta + a_0$, so that

$$(43) \quad \kappa_\lambda := \kappa - \lambda(a_0 - \alpha) \geq \eta \quad \text{for every } \lambda \in [0, 1],$$

the two displayed inequalities being the two endpoints of the affine function $\lambda \mapsto \kappa_\lambda$.

This is (30) with $8(L + \tilde{L})$ replaced by a_0 , and Section A.4 showed $a_0 = 8(L + \tilde{L})$ suffices when the context is overfitted; Section 6.5 gives a value valid in general.

6.2. The map $\mathcal{G}_\lambda$. The obstruction that Proposition 4.1 identifies is that a covariate $j \in \gamma^*$ may be worth adding in some contexts and not in others. We therefore split γ^* into three parts by comparing $\lambda \psi_j^\pm$ with $\kappa \log p$, leaving a margin $\log(2s^*)$ on each side.

DEFINITION 6.1 (Decided and undecided covariates). For $\lambda \in [0, 1]$ put

$$(44) \quad \begin{aligned} c_\lambda^- &:= \{j \in \gamma^* : \lambda \psi_j^- \geq \kappa \log p + \log(2s^*)\}, \\ c_\lambda^+ &:= \{j \in \gamma^* : \lambda \psi_j^+ > \kappa \log p - \log(2s^*)\}, \end{aligned} \quad C_\lambda := c_\lambda^+ \setminus c_\lambda^-, \quad k_\lambda := |C_\lambda|,$$

so that, for every $\lambda \geq 0$ and every $\gamma \subseteq \gamma^* \setminus \{j\}$,

$$(45) \quad j \in c_\lambda^- \implies \lambda \psi_j(\gamma) - \kappa \log p \geq \log(2s^*), \quad j \notin c_\lambda^+ \implies \lambda \psi_j(\gamma) - \kappa \log p \leq -\log(2s^*).$$

A covariate is certainly worth having when it beats the penalty in its *worst* context, whence ψ_j^- , and certainly not when it loses in its *best*, whence ψ_j^+ ; the margin $\log(2s^*)$ is what makes the drop in Theorem 6.1 $1/(2s^*)$ rather than 1, which is what pays for the branching in Lemma 6.2. From (44), with $\kappa \log p > \log(2s^*)$ by $\kappa \geq \eta$ and $p^{1-\eta} \leq \frac{1}{2}$,

$$(46) \quad c_\lambda^- \subseteq c_\lambda^+ \subseteq \gamma^*, \quad c_0^- = c_0^+ = \emptyset, \quad \lambda \mapsto c_\lambda^\mp \text{ non-decreasing:}$$

C_λ is a window sweeping γ^* as λ grows, and Remark 7.1 bounds how often it is non-empty.

DEFINITION 6.2 (The temperature-adapted map). For $\gamma \in \mathcal{M}(s_0)$ define $\mathcal{G}_\lambda(\gamma)$ by the first applicable rule:

- (H1) if $\gamma \not\subseteq \gamma^*$, delete $\max(\gamma \setminus \gamma^*)$;
- (H2) else if $c_\lambda^- \not\subseteq \gamma$, add $\min(c_\lambda^- \setminus \gamma)$;
- (H3) else if $\gamma \not\subseteq c_\lambda^+$, delete $\min(\gamma \setminus c_\lambda^+)$;
- (H4) else if $\gamma \cap C_\lambda \neq \emptyset$, delete $\min(\gamma \cap C_\lambda)$;
- (H5) else $\mathcal{G}_\lambda(\gamma) = \gamma$.

- LEMMA 6.1** ($\mathcal{G}_\lambda$ is a valid transition function). **(a)** Every rule changes one coordinate and keeps the state in $\mathcal{M}(s_0)$, so each edge it uses is a single flip; the exchange move of (12) is never needed.
- (b)** The unique fixed point of $\mathcal{G}_\lambda$ is c_λ^- , and $\mathcal{G}_\lambda^k(\gamma) = c_\lambda^-$ for some $k \leq 2s_0$.
- (c)** Along the orbit of any γ , the rules fire in the order (H1), then (H2) and (H3), then (H4): once a rule other than (H1) has fired, (H1) never fires again, and once (H4) has fired, neither (H2) nor (H3) fires again.

6.3. The drop bounds.

THEOREM 6.1 (Drop bound for $\mathcal{G}_\lambda$). Let $\lambda \in [0, 1]$, assume Assumption 6.1 with exponent η , and work on the good event. Let $\gamma \in \mathcal{M}(s_0) \setminus \{c_\lambda^-\}$. Then

$$(47) \quad \frac{\pi_n^{(\lambda)}(\gamma | Y)}{\pi_n^{(\lambda)}(\mathcal{G}_\lambda(\gamma) | Y)} \leq \begin{cases} p^{-\kappa_\lambda} \leq p^{-\eta} & \text{under rule (H1),} \\ \frac{1}{2s^*} & \text{under rules (H2) and (H3),} \\ \Theta_\lambda := \max_{j \in C_\lambda} \exp\{\lambda \psi_j^+ - \kappa \log p\} & \text{under rule (H4).} \end{cases}$$

Rules (H2) and (H3) are the point of the construction: they are uphill by the definition of $c_\lambda^\pm$. Only rule (H4) can downhill, and it fires at most k_λ times.

6.4. Precedents. Write $\Lambda_\lambda(\gamma)$ for the set of precedents of γ under $\mathcal{G}_\lambda$, defined as in (18) with $\mathcal{G}$ replaced by $\mathcal{G}_\lambda$.

LEMMA 6.2 (Counting). Let $\bar{\eta} \in \mathcal{M}(s_0)$. The number of $\bar{\gamma} \neq \bar{\eta}$ with $\mathcal{G}_\lambda(\bar{\gamma}) = \bar{\eta}$ is at most p if the rule applied to $\bar{\gamma}$ is (H1), at most $2s^*$ if it is (H2) or (H3), and at most k_λ if it is (H4).

COROLLARY 6.1 (Mass of the precedents, at every temperature). Let $\lambda \in [0, 1]$, assume Assumption 6.1 with an exponent η satisfying $p^{1-\eta} \leq \frac{1}{2}$, and work on the good event. Then for every $\gamma \in \mathcal{M}(s_0)$,

$$(48) \quad \frac{\pi_n^{(\lambda)}[\Lambda_\lambda(\gamma) | Y]}{\pi_n^{(\lambda)}(\gamma | Y)} \leq 2(2s^* + 1)(1 + k_\lambda \Theta_\lambda)^{k_\lambda}.$$

Note the contrast with Corollary 4.2: the condition on η is now $p^{1-\eta} \leq \frac{1}{2}$ rather than (31), because the branching of the influential steps is $2s^*$ rather than $(s^* + 1)p$ and is absorbed by the margin $\log(2s^*)$ built into (44) instead of by the drop $p^{-\eta}$.

6.5. *A sufficient a_0 .* In Section A.4 we bounded $\psi_j(\gamma)$ for $j \in S^c$ when $\gamma \supseteq \gamma^*$, where $(\Phi_{\gamma \cup \{j\}} - \Phi_\gamma)X_{\gamma^*}\beta_{\gamma^*}^* = 0$. When influential covariates are missing that term no longer vanishes, and what replaces it is $\omega(X)$, already present in Assumption 3.4.

PROPOSITION 6.1 (A value of a_0). *On the good event, and under (10), Assumption 6.1 holds with*

$$(49) \quad a_0 = 16(L + \tilde{L}) + \frac{8n\omega(X)\|\beta_{\gamma^*}^*\|_2^2}{\sigma_0^2 \log p}.$$

REMARK 6.1. In the regime of interest $n\|\beta_{\gamma^*}^*\|_2^2/\sigma_0^2 \asymp s^*nC_\beta^2/\sigma_0^2$, so the second term of (49) is of order $\omega(X)s^*nC_\beta^2/(\sigma_0^2 \log p)$ and Assumption 6.1 asks, in effect, for $\omega(X) \lesssim 1/s^*$. Assumption 3.4 already asks $\omega(X) \leq \nu^2(s_0 - s^*)/(4s^*)$, so this is a tightening of Assumption 3.4 by a factor of order $\nu^2 s_0$, not a hypothesis of a new kind. It is the one ingredient below that no choice of $\mathcal{G}_\lambda$ removes: if a covariate outside S reproduces the missing signal, then keeping it genuinely raises $\pi_n^{(\lambda)}(\cdot | Y)$, and that is a property of the target.

7. A gap at every temperature. The memoryless ensemble generated by $\mathcal{G}_\lambda$, built as in Section 3.3, would work; but rule (H4) would cost it a great deal. Following the proof of Lemma 5.1 with the drop bounds of Theorem 6.1 and the precedent bound of Corollary 6.1 gives a gap at least $[32ps_0(2s^* + 1)\max\{1, \Theta_\lambda\}(1 + k_\lambda\Theta_\lambda)^{k_\lambda}]^{-1}$, exponential in k_λ even when the undecided covariates are entirely harmless. They are harmless in a precise sense: $j \in C_\lambda$ means $\lambda\psi_j \approx \kappa \log p$, so including j is nearly neutral under $\pi_n^{(\lambda)}(\cdot | Y)$, and a nearly neutral coordinate is easy for the chain — it flips freely — not hard. What is hard is only that a memoryless ensemble must *commit* to a value for it.

This section removes the commitment. We keep rules (H1)–(H3), which leave a fixed point in every one of the 2^{k_λ} residual classes, and connect those classes by paths inside the hypercube they form, rather than contracting them to a single state. The price becomes the congestion of a k_λ -dimensional hypercube problem, which is 1 when the undecided covariates do not interact with each other.

7.1. *Blocks, funnels, and the induced measure.* Let F_λ be the map defined by rules (H1)–(H3) of Definition 6.2, with $F_\lambda(\gamma) = \gamma$ when none applies. For $J \subseteq C_\lambda$ put

$$(50) \quad B_J := \{\gamma \in \mathcal{M}(s_0) : \gamma \cap C_\lambda = J\}, \quad v_J := c_\lambda^- \cup J, \quad \bar{\pi}_\lambda(J) := \pi_n^{(\lambda)}(B_J | Y).$$

LEMMA 7.1 (The funnel). *Let $\lambda \in [0, 1]$ and let η satisfy $p^{1-\eta} \leq \frac{1}{2}$. For every $J \subseteq C_\lambda$:*

- (a) F_λ maps B_J into itself, and v_J is its unique fixed point there; consequently $B_J = \Lambda^{F_\lambda}(v_J)$ — precedents under F_λ , not under $\mathcal{G}_\lambda$ as in $\Lambda_\lambda(\cdot)$.
- (b) every step of F_λ is $\pi_n^{(\lambda)}$ -increasing, and the orbit of any $\gamma \in B_J$ has at most $2s_0$ edges, all single flips;
- (c) $\pi_n^{(\lambda)}(v_J | Y) \geq \bar{\pi}_\lambda(J)/\{2(2s^* + 1)\}$.

Part c is what makes a funnel admissible: routing all the traffic of a block through the single state v_J costs a factor $2(2s^* + 1)$ and no more, because v_J already carries that share of the block.

7.2. The ensemble and its gap.

DEFINITION 7.1 (Two-level ensemble). Identify 2^{C_λ} with $\{0, 1\}^{k_\lambda}$ by fixing the increasing order of C_λ . For $\gamma \in B_J$ and $\gamma' \in B_{J'}$ with $\gamma \neq \gamma'$, let $\tilde{T}_{\gamma, \gamma'}$ be the concatenation of

- (i) the F_λ -orbit from γ to v_J ;
- (ii) the path $v_J \rightarrow v_{J'}$ that flips the coordinates of $J \triangle J'$ in increasing order, all of whose states are of the form $v_{J''}$;
- (iii) the reverse of the F_λ -orbit from γ' to $v_{J'}$,

truncated at the first repeated state so that it is simple; and when $J = J'$ let $\tilde{T}_{\gamma, \gamma'}$ be the path of Section 5.1 built from F_λ inside B_J . Write $\tilde{T}_\lambda$ for the resulting ensemble.

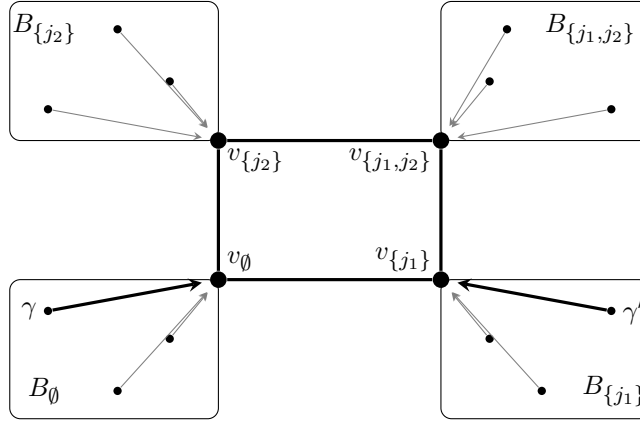


FIG 1. Definition 7.1 at $k_\lambda = 2$: arrows are steps of F_λ , funnelling each block B_J onto v_J ; the four v_J carry the hypercube $\{0, 1\}^{k_\lambda}$ of segment (ii).

See Fig. 1. Segment (ii) is legitimate: each $v_{J''}$ is a subset of γ^* , hence lies in $\mathcal{M}(s_0)$, and consecutive ones differ by one flip. The ensemble is *not* memoryless, which is why the precedent machinery of Section 6.4 applies to its first and third segments only.

DEFINITION 7.2 (Hypercube congestion). With $\bar{\pi}_\lambda$ as in (50), put

$$(51) \quad \varrho_\lambda := \max_{J, j \in C_\lambda} \frac{1}{\min\{\bar{\pi}_\lambda(J), \bar{\pi}_\lambda(J \triangle \{j\})\}} \sum_{(J_1, J_2)} \bar{\pi}_\lambda(J_1) \bar{\pi}_\lambda(J_2),$$

the inner sum being over the ordered pairs whose increasing-order path in $\{0, 1\}^{k_\lambda}$ traverses the directed edge $(J, J \triangle \{j\})$.

The congestion of the two-level ensemble now splits in two. Its first and third segments are the rules (H1)–(H3) of $\mathcal{G}_\lambda$, which Theorem 6.1 already controls; its middle segment lives inside a single hypercube, and everything it costs is ϱ_λ .

THEOREM 7.1 (Gap without commitment). Let $\lambda \in [0, 1]$, assume Assumption 6.1 with an exponent η satisfying $p^{1-\eta} \leq \frac{1}{2}$, and work on the good event. Then

$$(52) \quad \text{Gap}(\mathbf{P}_\lambda) \geq \frac{1}{40 p s_0 (2s^* + 1) \max\{1, \varrho_\lambda\}}.$$

COROLLARY 7.1 (A gap outside the windows). *Under the hypotheses of Theorem 7.1, if $k_\lambda = 0$ then*

$$(53) \quad \text{Gap}(\mathbf{P}_\lambda) \geq \frac{1}{40 p s_0 (2s^* + 1)}.$$

Indeed $C_\lambda = \emptyset$, so the maximum in (51) is over the empty set and $\max\{1, \varrho_\lambda\} = 1$; equivalently, there is a single block $B_\emptyset = \mathcal{M}(s_0)$ and $\tilde{T}_\lambda$ is the ensemble generated by F_λ . The bound is better than Lemma 5.1 by a factor of order s_0/s^* : $\mathcal{G}_\lambda$ clears the covariates outside γ^* before it touches the others and so never needs the exchange move, which is where the second factor s_0 came from. It holds in particular at $\lambda = 0$, where $c_0^- = c_0^+ = \emptyset$ and F_0 is the plain descent $\gamma \mapsto \gamma \setminus \max(\gamma)$.

REMARK 7.1 (How often is $k_\lambda > 0$). A covariate $j \in \gamma^*$ belongs to C_λ exactly when

$$\frac{\kappa \log p - \log(2s^*)}{\psi_j^+} < \lambda < \frac{\kappa \log p + \log(2s^*)}{\psi_j^-},$$

an interval of length at most $\kappa \log p (\psi_j^+ - \psi_j^-) / (\psi_j^- \psi_j^+) + \log(2s^*) \{1/\psi_j^- + 1/\psi_j^+\}$. Summing over $j \in \gamma^*$, the set $\{\lambda \in [0, 1] : k_\lambda \geq 1\}$ has Lebesgue measure at most that sum, which is $\mathcal{O}(s^* \log s^* / \min_j \psi_j^-)$ when the spreads $\psi_j^+ - \psi_j^-$ are small compared with the ψ_j^- . So Corollary 7.1 applies for all but a short list of temperature windows, whose endpoints are computable in advance; a tempering schedule can be chosen to step across them.

7.3. *When the undecided covariates do not interact.* It remains to bound ϱ_λ , and here the near-neutrality of C_λ enters. Everything depends on how far $\bar{\pi}_\lambda$ is from a product measure on $\{0, 1\}^{k_\lambda}$.

LEMMA 7.2 (Product measures are free). *If $\bar{\pi}_\lambda$ is a product measure on 2^{C_λ} , then $\varrho_\lambda \leq 1$.*

LEMMA 7.3 (Comparison). *Suppose there is a product measure $\bar{\mu}$ on 2^{C_λ} with $|\log\{\bar{\pi}_\lambda(J)/\bar{\mu}(J)\}| \leq D$ for every $J \subseteq C_\lambda$. Then $\varrho_\lambda \leq e^{3D}$.*

The deviation D is controlled by the interaction among the undecided covariates alone:

$$(54) \quad \Delta_\lambda^C := \max_{j \in C_\lambda} \max\{|\psi_j(c_\lambda^- \cup U) - \psi_j(c_\lambda^- \cup U')| : U, U' \subseteq C_\lambda \setminus \{j\}\}.$$

PROPOSITION 7.1 (Bound on ϱ_λ). *Under the hypotheses of Theorem 7.1,*

$$(55) \quad \varrho_\lambda \leq \{2(2s^* + 1)\}^6 \exp\{6k_\lambda \Delta_\lambda^C\}.$$

REMARK 7.2 (What the flow buys). Combining Theorem 7.1 and Proposition 7.1,

$$(56) \quad \text{Gap}(\mathbf{P}_\lambda) \gtrsim \frac{\exp\{-6k_\lambda \Delta_\lambda^C\}}{p s_0 (s^*)^7},$$

against the $(1 + k_\lambda \Theta_\lambda)^{k_\lambda}$ that the memoryless ensemble of Section 7 would have charged. The two differ in what the exponent contains. There it is $k_\lambda \log(k_\lambda \Theta_\lambda)$ with $\Theta_\lambda \geq (2s^*)^{-1}$ always, so the penalty is paid whenever $k_\lambda > 0$, however benign the undecided covariates. In (56) it is $6k_\lambda \Delta_\lambda^C$, which vanishes when the undecided covariates do not interact with each other, and then the bound is polynomial with no reference to k_λ at all.

This is the case that matters in practice. Undecided covariates are those with $\lambda \psi_j \approx \kappa \log p$, that is with nearly equal ψ_j ; by Lemma A.4a and the transfer between ψ_j and the explained

energy, nearly equal ψ_j means nearly equal $|\beta_j^*|$. So C_λ is typically a group of covariates with tied coefficients, and Δ_λ^C measures only how much those particular columns interact — a $k_\lambda \times k_\lambda$ question, not an s^* or s_0 one. For a design whose β^* takes two magnitudes, with multiplicities four and six, Δ_λ^C is the interaction among four or six independent Gaussian columns, which is $\mathcal{O}(1)$: a memoryless ensemble would charge $(1 + 6\Theta_\lambda)^6$ there, and Theorem 7.1 charges a constant.

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APPENDIX A: DEFERRED PROOFS

Every proof of the paper is collected here, together with the notation and the deterministic bounds they rest on. Only the proof of Theorem 4.1 involves the temperature, and it involves it only through the factor λ multiplying the fit term of (7).

A.1. Notation. We use throughout the effective noise of YWJ,

$$(57) \quad \tilde{w} := X_{S^c} \beta_{S^c}^* + w, \quad \text{so that} \quad Y = X_{\gamma^*} \beta_{\gamma^*}^* + \tilde{w},$$

the elementary inequalities

$$(58) \quad \log(1+x) \leq x \quad (x > -1), \quad \|a+b\|_2^2 \leq 2\|a\|_2^2 + 2\|b\|_2^2,$$

$$\frac{a}{a+b} \geq \min\left\{\frac{1}{2}, \frac{a}{2b}\right\} \quad (a, b > 0),$$

and, for an underfitted $\gamma \in \mathcal{M}(s_0)$, the quantity

$$(59) \quad A^2 := \nu \frac{\|(I - \Phi_\gamma) X_{\gamma^*} \beta_{\gamma^*}^*\|_2^2}{|\gamma^* \setminus \gamma|} > 0,$$

the dependence on γ being suppressed, as YWJ suppress it, since γ is fixed wherever A occurs.

A.2. Deterministic bounds.

LEMMA A.1 (Uniform residual lower bound). *On the good event, and under (10), every $\bar{\gamma} \in \mathcal{M}(s_0)$ satisfies*

$$(60) \quad \|Y\|_2^2 (g^{-1} + 1 - R_{\bar{\gamma}}^2) \geq \|(I - \Phi_{\bar{\gamma}})Y\|_2^2 \geq \frac{n\sigma_0^2}{8}.$$

PROOF. Since $\|Y\|_2^2 (g^{-1} + 1 - R_{\bar{\gamma}}^2) = g^{-1}\|Y\|_2^2 + \|(I - \Phi_{\bar{\gamma}})Y\|_2^2$, the first inequality amounts to discarding $g^{-1}\|Y\|_2^2 \geq 0$. For the second, fix $\bar{\gamma} \in \mathcal{M}(s_0)$ and set $\bar{\gamma}^+ := \bar{\gamma} \cup \gamma^*$, so that $|\bar{\gamma}^+| \leq |\bar{\gamma}| + s^* \leq s$ and $\bar{\gamma}^+ \in \mathcal{M}(s)$. Since $\bar{\gamma} \subseteq \bar{\gamma}^+$, the range of $\Phi_{\bar{\gamma}}$ is contained in that of $\Phi_{\bar{\gamma}^+}$, so $I - \Phi_{\bar{\gamma}} \succcurlyeq I - \Phi_{\bar{\gamma}^+}$ and

$$(61) \quad \|(I - \Phi_{\bar{\gamma}})Y\|_2^2 \geq \|(I - \Phi_{\bar{\gamma}^+})Y\|_2^2.$$

Because $\gamma^* \subseteq \bar{\gamma}^+$, the vector $X_{\gamma^*} \beta_{\gamma^*}^*$ lies in the range of $\Phi_{\bar{\gamma}^+}$, so $(I - \Phi_{\bar{\gamma}^+})X_{\gamma^*} \beta_{\gamma^*}^* = 0$ and

$$(I - \Phi_{\bar{\gamma}^+})Y = (I - \Phi_{\bar{\gamma}^+})w + (I - \Phi_{\bar{\gamma}^+})X_{S^c} \beta_{S^c}^*.$$

Apply $\|a + b\|_2^2 \geq \frac{1}{2}\|a\|_2^2 - \|b\|_2^2$, which is the rearrangement of $\frac{1}{2}(\|a\|_2 - 2\|b\|_2)^2 \geq 0$, with $a = (I - \Phi_{\bar{\gamma}+})w$ and $b = (I - \Phi_{\bar{\gamma}+})X_{S^c}\beta_{S^c}^*$. Since projections are non-expansive, $\|b\|_2^2 \leq \|X_{S^c}\beta_{S^c}^*\|_2^2 \leq \tilde{L}\sigma_0^2 \log p$ by Assumption 3.1, and $\|a\|_2^2 = \|w\|_2^2 - w^\top \Phi_{\bar{\gamma}+}w$ because $I - \Phi_{\bar{\gamma}+}$ is a projection. Hence

$$\|(I - \Phi_{\bar{\gamma}+})Y\|_2^2 \geq \frac{1}{2}(\|w\|_2^2 - w^\top \Phi_{\bar{\gamma}+}w) - \tilde{L}\sigma_0^2 \log p.$$

On $\mathcal{C}_n$ we have $\|w\|_2^2 \geq n\sigma_0^2/2$, and on $\mathcal{B}_n$, applied to $\bar{\gamma}^+ \in \mathcal{M}(s)$, we have $w^\top \Phi_{\bar{\gamma}+}w \leq 8s\sigma_0^2 \log p$. Therefore

$$\|(I - \Phi_{\bar{\gamma}+})Y\|_2^2 \geq \frac{n\sigma_0^2}{4} - \{4s + \tilde{L}\}\sigma_0^2 \log p \geq \frac{n\sigma_0^2}{4} - \{8s + \tilde{L}\}\sigma_0^2 \log p,$$

and by (10) the subtracted term is at most $n\sigma_0^2/32$, so the right-hand side is at least $7n\sigma_0^2/32 \geq n\sigma_0^2/8$. Combining with (61) finishes the proof. $\square$

LEMMA A.2 (Effective noise). *On the good event, and under (10):*

- (a) $\|\tilde{w}\|_2^2 \leq 2\tilde{L}\sigma_0^2 \log p + 3n\sigma_0^2$;
- (b) for every pair $\gamma_2 \subset \gamma_1$ of $\mathcal{M}(s)$ with $|\gamma_1| - |\gamma_2| = 1$,

$$\|(\Phi_{\gamma_1} - \Phi_{\gamma_2})\tilde{w}\|_2^2 \leq 2(L + \tilde{L})\sigma_0^2 \log p \leq \frac{n\nu^2 C_\beta^2}{64}.$$

PROOF. **a** By (58) and Assumption 3.1, $\|\tilde{w}\|_2^2 \leq 2\|X_{S^c}\beta_{S^c}^*\|_2^2 + 2\|w\|_2^2 \leq 2\tilde{L}\sigma_0^2 \log p + 2\|w\|_2^2$, and on $\mathcal{C}_n$ we have $\|w\|_2^2 \leq \frac{3}{2}n\sigma_0^2$.

b Write $\Pi = \Phi_{\gamma_1} - \Phi_{\gamma_2}$, an orthogonal projection, so that $\|\Pi v\|_2 \leq \|v\|_2$ for every v and $\|\Pi w\|_2^2 = w^\top \Pi w$. Hence, by (58),

$$\begin{aligned} \|\Pi \tilde{w}\|_2^2 &\leq 2\|\Pi X_{S^c}\beta_{S^c}^*\|_2^2 + 2\|\Pi w\|_2^2 \leq 2\|X_{S^c}\beta_{S^c}^*\|_2^2 + 2w^\top (\Phi_{\gamma_1} - \Phi_{\gamma_2})w \\ &\leq 2\tilde{L}\sigma_0^2 \log p + 2L\sigma_0^2 \log p, \end{aligned}$$

using Assumption 3.1 for the first term and the event $\mathcal{A}_n$, applied to the nested pair $\gamma_2 \subset \gamma_1$ of $\mathcal{M}(s)$ with $|\gamma_1| - |\gamma_2| = 1$, for the second. The final inequality is Assumption 3.5 rearranged. $\square$

LEMMA A.3 (Upper bound on the regularised residual). *On the good event, and under (10) and $n \geq 3$, every $\gamma \in \mathcal{M}(s_0)$ satisfies*

$$(62) \quad \|Y\|_2^2(g^{-1} + 1 - R_\gamma^2) \leq 2\|(I - \Phi_\gamma)X_{\gamma^*}\beta_{\gamma^*}^*\|_2^2 + 8n\sigma_0^2.$$

PROOF. By (57) and (58),

$$\|(I - \Phi_\gamma)Y\|_2^2 \leq 2\|(I - \Phi_\gamma)X_{\gamma^*}\beta_{\gamma^*}^*\|_2^2 + 2\|(I - \Phi_\gamma)\tilde{w}\|_2^2 \leq 2\|(I - \Phi_\gamma)X_{\gamma^*}\beta_{\gamma^*}^*\|_2^2 + 2\|\tilde{w}\|_2^2,$$

the last step because $I - \Phi_\gamma$ is a projection. By Lemma A.2a, $2\|\tilde{w}\|_2^2 \leq 4\tilde{L}\sigma_0^2 \log p + 6n\sigma_0^2$, and on $\mathcal{D}_n$ we have $g^{-1}\|Y\|_2^2 \leq (2\log p + 3)\sigma_0^2$. Adding the two, the additive term is at most

$$\{4\tilde{L}\log p + 6n + 2\log p + 3\}\sigma_0^2 \leq \left\{\frac{n}{8} + 6n + \frac{n}{128} + n\right\}\sigma_0^2 \leq 8n\sigma_0^2,$$

where we used, in order: $\tilde{L}\log p \leq n/32$ from (10), whence $4\tilde{L}\log p \leq n/8$; the same display with $\tilde{L}$ replaced by $8s \geq 8$, whence $\log p \leq n/256$ and $2\log p \leq n/128$; and $3 \leq n$. $\square$

LEMMA A.4 (Greedy selection; YWJ, Lemma 8). *Let $\gamma \in \mathcal{M}(s_0)$ be underfitted and let j_γ be as in (21). Under Assumptions 3.2, 3.4 and 3.5:*

(a) $\|(\Phi_{\gamma \cup \{j_\gamma\}} - \Phi_\gamma)X_{\gamma^*}\beta_{\gamma^*}^*\|_2^2 \geq A^2$, and moreover

$$(63) \quad A^2 \geq n\nu^2 \min_{j \in \gamma^*} |\beta_j^*|^2 \geq n\nu^2 C_\beta^2;$$

(b) if in addition γ is saturated and k_γ is as in (22), then, with $\tilde{\gamma} = \gamma \cup \{j_\gamma\}$ and $\gamma' = \tilde{\gamma} \setminus \{k_\gamma\}$,

$$\|(\Phi_{\tilde{\gamma}} - \Phi_{\gamma'})X_{\gamma^*}\beta_{\gamma^*}^*\|_2^2 \leq n\omega(X) \frac{\|\beta_{\gamma^* \setminus \gamma}^*\|_2^2}{s_0 - s^*} \leq \frac{A^2}{4}.$$

PROOF. We use two facts from YWJ. The first is their Lemma 7, the rank-one update formula: for $\bar{\gamma} \in \mathcal{M}(s)$ and $k \notin \bar{\gamma}$ with $|\bar{\gamma}| < s$,

$$(64) \quad \Phi_{\bar{\gamma} \cup \{k\}} - \Phi_{\bar{\gamma}} = \frac{(I - \Phi_{\bar{\gamma}})X_k X_k^\top (I - \Phi_{\bar{\gamma}})}{X_k^\top (I - \Phi_{\bar{\gamma}})X_k},$$

which follows from the block inversion formula applied to $X_{\bar{\gamma} \cup \{k\}}^\top X_{\bar{\gamma} \cup \{k\}}$. The second is their Lemma 6: under Assumption 3.2, for every nested pair $\gamma_2 \subset \gamma_1$ of $\mathcal{M}(s)$,

$$(65) \quad \lambda_{\min} \left(\frac{1}{n} X_{\gamma_1 \setminus \gamma_2}^\top (I - \Phi_{\gamma_2}) X_{\gamma_1 \setminus \gamma_2} \right) \geq \nu,$$

because the matrix inside is n times the Schur complement of the block $X_{\gamma_2}^\top X_{\gamma_2}$ in $X_{\gamma_1}^\top X_{\gamma_1}$, hence its inverse is the corresponding diagonal block of $(n^{-1} X_{\gamma_1}^\top X_{\gamma_1})^{-1}$, whose largest eigenvalue is at most $\lambda_{\min}(n^{-1} X_{\gamma_1}^\top X_{\gamma_1})^{-1} \leq \nu^{-1}$.

Proof of a. Put $u := (I - \Phi_\gamma)X_{\gamma^*}\beta_{\gamma^*}^*$. For $\ell \in \gamma^* \setminus \gamma$, idempotence of projections and (64) give

$$(66) \quad \begin{aligned} \|\Phi_{\gamma \cup \{\ell\}}X_{\gamma^*}\beta_{\gamma^*}^*\|_2^2 - \|\Phi_\gamma X_{\gamma^*}\beta_{\gamma^*}^*\|_2^2 &= (\beta_{\gamma^*}^*)^\top X_{\gamma^*}^\top (\Phi_{\gamma \cup \{\ell\}} - \Phi_\gamma) X_{\gamma^*}\beta_{\gamma^*}^* \\ &= \frac{\langle X_\ell, u \rangle^2}{X_\ell^\top (I - \Phi_\gamma) X_\ell} \geq \frac{\langle X_\ell, u \rangle^2}{n}, \end{aligned}$$

where the last step uses $X_\ell^\top (I - \Phi_\gamma) X_\ell \leq \|X_\ell\|_2^2 = n$ and, for the middle equality, that $X_\ell^\top (I - \Phi_\gamma) X_{\gamma^*}\beta_{\gamma^*}^* = \langle X_\ell, u \rangle$. Summing (66) over $\ell \in \gamma^* \setminus \gamma$,

$$(67) \quad \sum_{\ell \in \gamma^* \setminus \gamma} \left\{ \|\Phi_{\gamma \cup \{\ell\}}X_{\gamma^*}\beta_{\gamma^*}^*\|_2^2 - \|\Phi_\gamma X_{\gamma^*}\beta_{\gamma^*}^*\|_2^2 \right\} \geq \frac{1}{n} \|X_{\gamma^* \setminus \gamma}^\top u\|_2^2 = \frac{1}{n} \|X_{\gamma^* \cup \gamma}^\top u\|_2^2,$$

the last equality because $X_j^\top u = 0$ for $j \in \gamma$: indeed X_j lies in the range of Φ_γ , to which u is orthogonal by construction. Now u lies in the range of $X_{\gamma^* \cup \gamma}$, so $u = X_{\gamma^* \cup \gamma} v$ for some v ; writing $M := X_{\gamma^* \cup \gamma}^\top X_{\gamma^* \cup \gamma}$, we have $|\gamma^* \cup \gamma| \leq s$, hence $M \succcurlyeq n\nu I$ by Assumption 3.2, and therefore

$$\frac{1}{n} \|X_{\gamma^* \cup \gamma}^\top u\|_2^2 = \frac{1}{n} v^\top M^2 v \geq \nu v^\top M v = \nu \|u\|_2^2,$$

the inequality because $M^2 - n\nu M = M^{1/2}(M - n\nu I)M^{1/2} \succcurlyeq 0$. Combining with (67) and dividing by $|\gamma^* \setminus \gamma|$, the maximum over ℓ of the left-hand side of (66) is at least $\nu \|u\|_2^2 / |\gamma^* \setminus \gamma| = A^2$. Since j_γ attains that maximum by (21), and since the left-hand side of (66) equals $\|(\Phi_{\gamma \cup \{j_\gamma\}} - \Phi_\gamma)X_{\gamma^*}\beta_{\gamma^*}^*\|_2^2$ by idempotence, the first claim follows.

For (63), note that $(I - \Phi_\gamma)X_{\gamma^*}\beta_{\gamma^*}^* = (I - \Phi_\gamma)X_{\gamma^* \setminus \gamma}\beta_{\gamma^* \setminus \gamma}^*$, because the columns X_j with $j \in \gamma^* \cap \gamma$ lie in the range of Φ_γ ; hence by (65) applied to $\gamma \subset \gamma^* \cup \gamma$,

$$(68) \quad \|u\|_2^2 = (\beta_{\gamma^* \setminus \gamma}^*)^\top X_{\gamma^* \setminus \gamma}^\top (I - \Phi_\gamma) X_{\gamma^* \setminus \gamma} \beta_{\gamma^* \setminus \gamma}^* \geq n\nu \|\beta_{\gamma^* \setminus \gamma}^*\|_2^2,$$

so that, using $\|\beta_{\gamma^* \setminus \gamma}^*\|_2^2 \geq |\gamma^* \setminus \gamma| \min_{j \in \gamma^*} |\beta_j^*|^2$ and (3),

$$(69) \quad A^2 = \frac{\nu \|u\|_2^2}{|\gamma^* \setminus \gamma|} \geq \frac{n\nu^2 \|\beta_{\gamma^* \setminus \gamma}^*\|_2^2}{|\gamma^* \setminus \gamma|} \geq n\nu^2 \min_{j \in \gamma^*} |\beta_j^*|^2 \geq n\nu^2 C_\beta^2.$$

Proof of b. Write $\tilde{\gamma} = \gamma \cup \{j_\gamma\}$. For $\tilde{\gamma} \in \mathcal{M}(s)$ let $\hat{\beta}(\tilde{\gamma})$ solve

$$(70) \quad \min \{ \|X\beta - X_{\gamma^* \setminus \tilde{\gamma}} \beta_{\gamma^* \setminus \tilde{\gamma}}^*\|_2^2 : \beta \in \mathbb{R}^p, \beta_j = 0 \text{ for } j \notin \tilde{\gamma} \},$$

so that

$$(71) \quad \|X\hat{\beta}(\tilde{\gamma}) - X_{\gamma^* \setminus \tilde{\gamma}} \beta_{\gamma^* \setminus \tilde{\gamma}}^*\|_2^2 = \|(I - \Phi_{\tilde{\gamma}})X_{\gamma^* \setminus \tilde{\gamma}} \beta_{\gamma^* \setminus \tilde{\gamma}}^*\|_2^2, \quad \hat{\beta}(\tilde{\gamma})_{\tilde{\gamma}} = (X_{\tilde{\gamma}}^\top X_{\tilde{\gamma}})^{-1} X_{\tilde{\gamma}}^\top X_{\gamma^* \setminus \tilde{\gamma}} \beta_{\gamma^* \setminus \tilde{\gamma}}^*.$$

Since $k_\gamma \notin \gamma^*$ we have $\Phi_{\tilde{\gamma}} X_{\gamma^* \cap \tilde{\gamma}} = \Phi_{\tilde{\gamma} \setminus \{k\}} X_{\gamma^* \cap \tilde{\gamma}}$ for every $k \in \tilde{\gamma} \setminus \gamma^*$, so with $\gamma' = \tilde{\gamma} \setminus \{k_\gamma\}$

$$(72) \quad \begin{aligned} \|(\Phi_{\tilde{\gamma}} - \Phi_{\gamma'})X_{\gamma^*} \beta_{\gamma^*}^*\|_2^2 &= \|(\Phi_{\tilde{\gamma}} - \Phi_{\gamma'})X_{\gamma^* \setminus \tilde{\gamma}} \beta_{\gamma^* \setminus \tilde{\gamma}}^*\|_2^2 \\ &= \|(I - \Phi_{\gamma'})X_{\gamma^* \setminus \tilde{\gamma}} \beta_{\gamma^* \setminus \tilde{\gamma}}^*\|_2^2 - \|(I - \Phi_{\tilde{\gamma}})X_{\gamma^* \setminus \tilde{\gamma}} \beta_{\gamma^* \setminus \tilde{\gamma}}^*\|_2^2 \\ &= \|X\hat{\beta}(\gamma') - X_{\gamma^* \setminus \tilde{\gamma}} \beta_{\gamma^* \setminus \tilde{\gamma}}^*\|_2^2 - \|X\hat{\beta}(\tilde{\gamma}) - X_{\gamma^* \setminus \tilde{\gamma}} \beta_{\gamma^* \setminus \tilde{\gamma}}^*\|_2^2, \end{aligned}$$

using in the second line that $\Phi_{\tilde{\gamma}} - \Phi_{\gamma'}$ is a projection and $\gamma' \subset \tilde{\gamma}$. By optimality of $\hat{\beta}(\tilde{\gamma})$ in (70), the residual $X\hat{\beta}(\tilde{\gamma}) - X_{\gamma^* \setminus \tilde{\gamma}} \beta_{\gamma^* \setminus \tilde{\gamma}}^*$ is orthogonal to X_k for every $k \in \tilde{\gamma}$, so for such k

$$\begin{aligned} \|X\hat{\beta}(\tilde{\gamma} \setminus \{k\}) - X_{\gamma^* \setminus \tilde{\gamma}} \beta_{\gamma^* \setminus \tilde{\gamma}}^*\|_2^2 &\leq \|X\hat{\beta}(\tilde{\gamma}) - X_{\gamma^* \setminus \tilde{\gamma}} \beta_{\gamma^* \setminus \tilde{\gamma}}^* - X_k \hat{\beta}_k(\tilde{\gamma})\|_2^2 \\ &= \|X\hat{\beta}(\tilde{\gamma}) - X_{\gamma^* \setminus \tilde{\gamma}} \beta_{\gamma^* \setminus \tilde{\gamma}}^*\|_2^2 + \|X_k\|_2^2 |\hat{\beta}_k(\tilde{\gamma})|^2, \end{aligned}$$

the first line because the vector $\hat{\beta}(\tilde{\gamma})$ with its k th coordinate zeroed is feasible for the problem defining $\hat{\beta}(\tilde{\gamma} \setminus \{k\})$, and the second by orthogonality. Since $\|X_k\|_2^2 = n$ and k_γ minimises the left-hand side over $k \in \gamma \setminus \gamma^* = \tilde{\gamma} \setminus (\gamma^* \cup \{j_\gamma\})$, whose cardinality is $s_0 - |\gamma \cap \gamma^*| \geq s_0 - s^*$,

$$\|X\hat{\beta}(\gamma') - X_{\gamma^* \setminus \tilde{\gamma}} \beta_{\gamma^* \setminus \tilde{\gamma}}^*\|_2^2 - \|X\hat{\beta}(\tilde{\gamma}) - X_{\gamma^* \setminus \tilde{\gamma}} \beta_{\gamma^* \setminus \tilde{\gamma}}^*\|_2^2 \leq n \min_{k \in \gamma \setminus \gamma^*} |\hat{\beta}_k(\tilde{\gamma})|^2 \leq \frac{n \|\hat{\beta}(\tilde{\gamma})\|_2^2}{s_0 - s^*},$$

a minimum being at most the average. Finally

$$\|\hat{\beta}(\tilde{\gamma})\|_2^2 = \|(X_{\tilde{\gamma}}^\top X_{\tilde{\gamma}})^{-1} X_{\tilde{\gamma}}^\top X_{\gamma^* \setminus \tilde{\gamma}} \beta_{\gamma^* \setminus \tilde{\gamma}}^*\|_2^2 \leq \omega(X) \|\beta_{\gamma^* \setminus \tilde{\gamma}}^*\|_2^2 \leq \omega(X) \|\beta_{\gamma^* \setminus \gamma}^*\|_2^2,$$

by the definition of $\omega(X)$ in Assumption 3.4 and $\gamma^* \setminus \tilde{\gamma} \subseteq \gamma^* \setminus \gamma$. Together with (72) this gives the first inequality of b.

For the second, combine $A^2 \geq n\nu^2 \|\beta_{\gamma^* \setminus \gamma}^*\|_2^2 / |\gamma^* \setminus \gamma| \geq n\nu^2 \|\beta_{\gamma^* \setminus \gamma}^*\|_2^2 / s^*$, which is (69), with the first inequality of Assumption 3.4: $s_0 \geq (4\nu^{-2}\omega(X) + 1)s^*$ is equivalent to $4\omega(X)s^* \leq \nu^2(s_0 - s^*)$, that is to

$$\frac{n\omega(X) \|\beta_{\gamma^* \setminus \gamma}^*\|_2^2}{s_0 - s^*} \leq \frac{1}{4} \cdot \frac{n\nu^2 \|\beta_{\gamma^* \setminus \gamma}^*\|_2^2}{s^*} \leq \frac{A^2}{4}. \quad \square$$

REMARK A.1. With their constant 2 in place of our 4 in (10), YWJ obtain $\|v_2\|_2 \leq A/\sqrt{2}$ rather than $\|v_2\|_2 \leq A/2$, and combine it with a noise bound $A/4$. The resulting numbers do not close: on their page 2527 the displayed lower bound $A(A - A/4) - (A/\sqrt{2})(A/\sqrt{2} + A/4) - A^2/16$ equals $0.0107A^2$, not the $A^2/8$ claimed, and it is negative if one keeps the factor 2 that their own preceding display carries on the cross terms. Strengthening Assumption 3.4 to $s_0 \geq (4\nu^{-2}\omega(X) + 1)s^*$, and Assumption 3.5 so that the noise budget is $A/8$ rather than $A/4$, repairs this with room to spare; see (92) below. Nothing else in their argument, or in ours, is affected.

LEMMA A.5 (Proxy bound). *For every $\gamma \in \mathcal{M}(s)$ and every $j \in \gamma \setminus \gamma^*$,*

$$(73) \quad \|(\Phi_\gamma - \Phi_{\gamma \setminus \{j\}})X_{\gamma^* \setminus \gamma} \beta_{\gamma^*}^*\|_2^2 \leq n \omega(X) \|\beta_{\gamma^* \setminus \gamma}^*\|_2^2.$$

PROOF. This is the chain of displays in the proof of Lemma A.4b, run with $\tilde{\gamma}$ replaced by γ and k_γ by j . Every step used only $j \notin \gamma^*$ and $\gamma \in \mathcal{M}(s)$: first $\Phi_\gamma X_{\gamma^* \cap \gamma} = \Phi_{\gamma \setminus \{j\}} X_{\gamma^* \cap \gamma}$, so the left side equals $\|(\Phi_\gamma - \Phi_{\gamma \setminus \{j\}})X_{\gamma^* \setminus \gamma} \beta_{\gamma^*}^*\|_2^2$; then the identification with a difference of least-squares residuals; then feasibility of zeroing the j th coordinate, which gives the bound $n|\hat{\beta}_j(\gamma)|^2$; then $\|\hat{\beta}(\gamma)\|_2^2 \leq \omega(X) \|\beta_{\gamma^* \setminus \gamma}^*\|_2^2$. Taking $\gamma \supseteq \gamma^*$ makes the right side vanish and recovers the identity used in Section A.4. $\square$

A.3. Counting and paths.

LEMMA A.6 (Counting precedents). *Let $\bar{\eta} \in \mathcal{M}(s_0)$.*

- (a) $\mathcal{G}$ maps overfitted states to states containing γ^* . Consequently, if γ is underfitted then every $\bar{\gamma} \in \Lambda(\gamma)$ is underfitted.
- (b) $\bar{\eta}$ has at most $(s^* + 1)p$ immediate precedents that are underfitted — that is, states $\bar{\gamma} \neq \bar{\eta}$ with $\mathcal{G}(\bar{\gamma}) = \bar{\eta}$ and $\bar{\gamma}$ underfitted — and at most p that are overfitted.
- (c) Hence, for every $k \geq 0$, the number of $\bar{\gamma}$ with $\mathcal{G}^k(\bar{\gamma}) = \bar{\eta}$ and $\bar{\gamma}, \mathcal{G}(\bar{\gamma}), \dots, \mathcal{G}^{k-1}(\bar{\gamma})$ all underfitted is at most $\{(s^* + 1)p\}^k$; and if $\bar{\eta} \supseteq \gamma^*$, the number of overfitted $\bar{\gamma} \in \Lambda_k(\bar{\eta})$ is at most p^k .

PROOF. **a** An overfitted $\bar{\gamma}$ is treated by rule (G1), which deletes $\ell_{\bar{\gamma}} \in \bar{\gamma} \setminus \gamma^*$ and therefore leaves $\gamma^* \subseteq \mathcal{G}(\bar{\gamma})$; and γ^* is a fixed point. So $\mathcal{G}^k(\bar{\gamma}) \supseteq \gamma^*$ for every $k \geq 0$, and no such state is underfitted.

b Let $\bar{\gamma} \neq \bar{\eta}$ be underfitted with $\mathcal{G}(\bar{\gamma}) = \bar{\eta}$. If rule (G2) applies then $\bar{\eta} = \bar{\gamma} \cup \{j_{\bar{\gamma}}\}$ with $j_{\bar{\gamma}} \in \gamma^*$, so $\bar{\gamma} = \bar{\eta} \setminus \{j\}$ for some $j \in \gamma^* \cap \bar{\eta}$: at most s^* states. If rule (G3) applies then $\bar{\eta} = \bar{\gamma} \cup \{j_{\bar{\gamma}}\} \setminus \{k_{\bar{\gamma}}\}$ with $j_{\bar{\gamma}} \in \gamma^*$ and $k_{\bar{\gamma}} \in S^c$, so $\bar{\gamma} = \bar{\eta} \cup \{k\} \setminus \{j\}$ with $j \in \gamma^* \cap \bar{\eta}$ and $k \in S^c$: at most $s^*(p - s^*)$ states. The total is at most $s^*(1 + p - s^*) \leq s^*(1 + p) \leq (s^* + 1)p$, the last step because $s^* \leq p$. If instead $\bar{\gamma}$ is overfitted then rule (G1) applies and $\bar{\gamma} = \bar{\eta} \cup \{\ell\}$ with $\ell \in S^c$: at most p states.

c Iterate **b**: a state counted at level k is an immediate precedent, of the stated type, of one counted at level $k - 1$. For the second claim, **a** shows that an overfitted $\bar{\gamma} \in \Lambda_k(\bar{\eta})$ reaches $\bar{\eta}$ through states containing γ^* only, so each of the k steps deletes an index of S^c ; hence $\bar{\gamma} = \bar{\eta} \cup u$ with $u \subseteq S^c$ and $|u| = k$, and there are at most $\binom{p}{k} \leq p^k$ such u . $\square$

- LEMMA A.7 (Path properties; YWJ, Lemma 2). (a) $\mathcal{G}^k(\gamma) = \gamma^*$ for some $k \leq d_H(\gamma, \gamma^*)$, every $\gamma \in \mathcal{M}(s_0)$; in particular $\Lambda(\gamma^*) = \mathcal{M}(s_0)$.
- (b) $|T_{\gamma, \gamma^*}| \leq d_H(\gamma, \gamma^*) \leq s_0 + s^* \leq 2s_0$, and $\ell(\mathcal{T}) \leq 4s_0$.
 - (c) For a directed edge $e = (\bar{\gamma}, \mathcal{G}(\bar{\gamma}))$,

$$\{(\gamma, \gamma') : T_{\gamma, \gamma'} \ni e\} \subseteq \Lambda(\bar{\gamma}) \times \mathcal{M}(s_0).$$

PROOF. **a** By the second remark after Definition 3.3, $d_H(\mathcal{G}(\gamma), \gamma^*) < d_H(\gamma, \gamma^*)$ whenever $\gamma \neq \gamma^*$, so the orbit reaches the unique fixed point γ^* in at most $d_H(\gamma, \gamma^*)$ steps. Hence every state lies in $\Lambda(\gamma^*)$.

b Each step of $\mathcal{G}$ decreases $d_H(\cdot, \gamma^*)$ by at least one, so the orbit has at most $d_H(\gamma, \gamma^*)$ edges; and $d_H(\gamma, \gamma^*) \leq |\gamma| + |\gamma^*| \leq s_0 + s^*$, which is at most $2s_0$ because $s^* < s_0$. Since T_{γ, γ^*} is contained in the concatenation of T_{γ, γ^*} with the reverse of T_{γ', γ^*} , its length is at most $4s_0$.

c By construction $T_{\gamma, \gamma'}$ traverses its first segment in the direction of $\mathcal{G}$ and its second segment against it. The edge e points in the direction of $\mathcal{G}$, so it can only occur in the first segment, which is part of T_{γ, γ^*} . Hence $\bar{\gamma} \in T_{\gamma, \gamma^*}$, that is $\gamma \in \Lambda(\bar{\gamma})$, while γ' is unconstrained. $\square$

A.4. Proof of Theorem 4.1: overfitted states. Throughout Sections A.4 to A.6 we write $\gamma' := \mathcal{G}(\gamma)$ and start from (7) with the size change read off from (23).

Here $\gamma \supseteq \gamma^*$, $\gamma \neq \gamma^*$, and rule (G1) applies: $\gamma' = \gamma \setminus \{\ell_\gamma\}$ with $\ell_\gamma \in \gamma \setminus \gamma^*$, so $|\gamma'| - |\gamma| = -1$ and (7) gives

$$(74) \quad \log \frac{\pi_n^{(\lambda)}(\gamma | Y)}{\pi_n^{(\lambda)}(\gamma' | Y)} = -\kappa \log p - \frac{\lambda}{2} \log(1+g) + \frac{\lambda n}{2} \log \frac{g^{-1} + 1 - R_{\gamma'}^2}{g^{-1} + 1 - R_\gamma^2}.$$

Since $\gamma' \subset \gamma$ and $|\gamma| - |\gamma'| = 1$, the matrix $\Phi_\gamma - \Phi_{\gamma'}$ is an orthogonal projection of rank one, and

$$(75) \quad \|Y\|_2^2 (R_\gamma^2 - R_{\gamma'}^2) = Y^\top (\Phi_\gamma - \Phi_{\gamma'}) Y = \|(\Phi_\gamma - \Phi_{\gamma'}) Y\|_2^2 \geq 0,$$

so that, using $\log(1+x) \leq x$ and then Lemma A.1,

$$(76) \quad \frac{n}{2} \log \frac{g^{-1} + 1 - R_{\gamma'}^2}{g^{-1} + 1 - R_\gamma^2} = \frac{n}{2} \log \left(1 + \frac{\|(\Phi_\gamma - \Phi_{\gamma'}) Y\|_2^2}{\|Y\|_2^2 (g^{-1} + 1 - R_\gamma^2)} \right) \leq \frac{n}{2} \cdot \frac{8 \|(\Phi_\gamma - \Phi_{\gamma'}) Y\|_2^2}{n \sigma_0^2}.$$

It remains to bound $\|(\Phi_\gamma - \Phi_{\gamma'}) Y\|_2^2$. Since $\ell_\gamma \notin \gamma^*$ and $\gamma^* \subseteq \gamma$, we have $\gamma^* \subseteq \gamma'$, so $X_{\gamma^*} \beta_{\gamma^*}$ lies in the range of $\Phi_{\gamma'}$; and $\Phi_\gamma - \Phi_{\gamma'}$ annihilates that range, because for v in it $\Phi_\gamma v = v$, the range of $\Phi_{\gamma'}$ being contained in that of Φ_γ , so that $(\Phi_\gamma - \Phi_{\gamma'}) v = v - v = 0$. Hence, by (57),

$$(77) \quad (\Phi_\gamma - \Phi_{\gamma'}) Y = (\Phi_\gamma - \Phi_{\gamma'}) \tilde{w}, \quad \text{so} \quad \|(\Phi_\gamma - \Phi_{\gamma'}) Y\|_2^2 \leq 2(L + \tilde{L}) \sigma_0^2 \log p$$

by Lemma A.2b, applied to the nested pair $\gamma' \subset \gamma$ of $\mathcal{M}(s_0) \subseteq \mathcal{M}(s)$ with $|\gamma| - |\gamma'| = 1$. Substituting (77) into (76),

$$(78) \quad \frac{n}{2} \log \frac{g^{-1} + 1 - R_{\gamma'}^2}{g^{-1} + 1 - R_\gamma^2} \leq 8(L + \tilde{L}) \log p,$$

and substituting this into (74), using $\lambda \geq 0$ so that multiplying (78) by λ preserves the inequality,

$$\log \frac{\pi_n^{(\lambda)}(\gamma | Y)}{\pi_n^{(\lambda)}(\gamma' | Y)} \leq -\kappa \log p - \lambda \left\{ \frac{1}{2} \log(1+g) - 8(L + \tilde{L}) \log p \right\},$$

which is (25). $\square$

A.5. Proof of Theorem 4.1: underfitted, unsaturated states. Here $\gamma^* \setminus \gamma \neq \emptyset$ and $|\gamma| < s_0$, so rule (G2) applies: $\gamma' = \gamma \cup \{j_\gamma\}$, $|\gamma'| - |\gamma| = +1$, and (7) gives

$$(79) \quad \log \frac{\pi_n^{(\lambda)}(\gamma | Y)}{\pi_n^{(\lambda)}(\gamma' | Y)} = \kappa \log p + \frac{\lambda}{2} \log(1+g) + \frac{\lambda n}{2} \log \frac{g^{-1} + 1 - R_{\gamma'}^2}{g^{-1} + 1 - R_\gamma^2}.$$

Note $s^* \geq 1$, since $\gamma^* \setminus \gamma \neq \emptyset$. The matrix $\Phi_{\gamma'} - \Phi_\gamma$ is an orthogonal projection of rank one, and as in (75),

$$(80) \quad \|Y\|_2^2 (R_{\gamma'}^2 - R_\gamma^2) = \|(\Phi_{\gamma'} - \Phi_\gamma) Y\|_2^2,$$

so that, by $\log(1+x) \leq x$ applied with $x = -\|(\Phi_{\gamma'} - \Phi_\gamma)Y\|_2^2 / \{\|Y\|_2^2(g^{-1} + 1 - R_\gamma^2)\} \in (-1, 0]$,

$$(81) \quad \frac{n}{2} \log \frac{g^{-1} + 1 - R_{\gamma'}^2}{g^{-1} + 1 - R_\gamma^2} \leq -\frac{n}{2} \cdot \frac{\|(\Phi_{\gamma'} - \Phi_\gamma)Y\|_2^2}{\|Y\|_2^2(g^{-1} + 1 - R_\gamma^2)}.$$

Step 1: the numerator. By Lemma A.4a, $\|(\Phi_{\gamma'} - \Phi_\gamma)X_{\gamma^*}\beta_{\gamma^*}^*\|_2 \geq A$, and by Lemma A.2b applied to the nested pair $\gamma \subset \gamma'$, together with (63),

$$(82) \quad \|(\Phi_{\gamma'} - \Phi_\gamma)\tilde{w}\|_2^2 \leq \frac{n\nu^2 C_\beta^2}{64} \leq \frac{A^2}{64}, \quad \text{that is} \quad \|(\Phi_{\gamma'} - \Phi_\gamma)\tilde{w}\|_2 \leq \frac{A}{8}.$$

Hence, by the triangle inequality and (57),

$$(83) \quad \|(\Phi_{\gamma'} - \Phi_\gamma)Y\|_2 \geq A - \frac{A}{8} = \frac{7A}{8} > 0, \quad \text{so} \quad \|(\Phi_{\gamma'} - \Phi_\gamma)Y\|_2^2 \geq \frac{49}{64}A^2 \geq \frac{1}{2}A^2.$$

Step 2: the denominator. By Lemma A.3 and the definition (59), which gives $\|(I - \Phi_\gamma)X_{\gamma^*}\beta_{\gamma^*}^*\|_2^2 = |\gamma^* \setminus \gamma|A^2/\nu \leq s^*A^2/\nu$,

$$(84) \quad \|Y\|_2^2(g^{-1} + 1 - R_\gamma^2) \leq \frac{2s^*A^2}{\nu} + 8n\sigma_0^2.$$

Step 3: combining. By (83) and (84), then by the third inequality of (58) with $a = 2s^*A^2/\nu > 0$ and $b = 8n\sigma_0^2 > 0$,

$$(85) \quad \frac{n}{2} \cdot \frac{\|(\Phi_{\gamma'} - \Phi_\gamma)Y\|_2^2}{\|Y\|_2^2(g^{-1} + 1 - R_\gamma^2)} \geq \frac{n}{4} \cdot \frac{A^2}{2s^*A^2/\nu + 8n\sigma_0^2} = \frac{n\nu}{8s^*} \cdot \frac{a}{a+b} \geq \frac{n\nu}{8s^*} \min\left\{\frac{1}{2}, \frac{s^*A^2}{8\nu n\sigma_0^2}\right\},$$

that is, distributing $n\nu/(8s^*)$ over the minimum and then using $A^2 \geq n\nu^2 C_\beta^2$ from (63),

$$(86) \quad \frac{n}{2} \cdot \frac{\|(\Phi_{\gamma'} - \Phi_\gamma)Y\|_2^2}{\|Y\|_2^2(g^{-1} + 1 - R_\gamma^2)} \geq \min\left\{\frac{n\nu}{16s^*}, \frac{A^2}{64\sigma_0^2}\right\} \geq \min\left\{\frac{n\nu}{16s^*}, \frac{n\nu^2 C_\beta^2}{64\sigma_0^2}\right\}.$$

Combining (79), (81) and (86), and using $\lambda \geq 0$,

$$\log \frac{\pi_n^{(\lambda)}(\gamma | Y)}{\pi_n^{(\lambda)}(\gamma' | Y)} \leq \kappa \log p + \lambda \left\{ \frac{1}{2} \log(1+g) - \min\left\{\frac{n\nu}{16s^*}, \frac{n\nu^2 C_\beta^2}{64\sigma_0^2}\right\} \right\},$$

which is (26). $\square$

A.6. Proof of Theorem 4.1: underfitted, saturated states. Here $\gamma^* \setminus \gamma \neq \emptyset$ and $|\gamma| = s_0$, so rule (G3) applies: with $\tilde{\gamma} = \gamma \cup \{j_\gamma\}$ and $\gamma' = \tilde{\gamma} \setminus \{k_\gamma\}$ we have $|\gamma'| = |\gamma|$, and (7) gives

$$(87) \quad \log \frac{\pi_n^{(\lambda)}(\gamma | Y)}{\pi_n^{(\lambda)}(\gamma' | Y)} = \frac{\lambda n}{2} \log \frac{g^{-1} + 1 - R_{\gamma'}^2}{g^{-1} + 1 - R_\gamma^2},$$

the inclusion penalty having cancelled identically. Note that $\tilde{\gamma} \in \mathcal{M}(s)$, since $|\tilde{\gamma}| = s_0 + 1 \leq s$, and that both γ and γ' are subsets of $\tilde{\gamma}$ of cardinality $|\tilde{\gamma}| - 1$, so that $\Phi_{\tilde{\gamma}} - \Phi_\gamma$ and $\Phi_{\tilde{\gamma}} - \Phi_{\gamma'}$ are orthogonal projections of rank one. Following YWJ, set

$$(88) \quad v_1 := (\Phi_{\tilde{\gamma}} - \Phi_\gamma)X_{\gamma^*}\beta_{\gamma^*}^*, \quad v_2 := (\Phi_{\tilde{\gamma}} - \Phi_{\gamma'})X_{\gamma^*}\beta_{\gamma^*}^*.$$

Adding and subtracting $Y^\top \Phi_{\tilde{\gamma}} Y$,

$$(89) \quad \|Y\|_2^2(R_{\gamma'}^2 - R_\gamma^2) = Y^\top \Phi_{\gamma'} Y - Y^\top \Phi_\gamma Y = \|(\Phi_{\tilde{\gamma}} - \Phi_\gamma)Y\|_2^2 - \|(\Phi_{\tilde{\gamma}} - \Phi_{\gamma'})Y\|_2^2.$$

By Lemma A.4a and b, $\|v_1\|_2 \geq A$ and $\|v_2\|_2 \leq A/2$; and by Lemma A.2b applied to each of the two nested pairs $\gamma \subset \tilde{\gamma}$ and $\gamma' \subset \tilde{\gamma}$ of $\mathcal{M}(s)$, exactly as in (82),

$$(90) \quad \|(\Phi_{\tilde{\gamma}} - \Phi_{\gamma})\tilde{w}\|_2 \leq \frac{A}{8}, \quad \|(\Phi_{\tilde{\gamma}} - \Phi_{\gamma'})\tilde{w}\|_2 \leq \frac{A}{8}.$$

Therefore, by the triangle inequality applied to $(\Phi_{\tilde{\gamma}} - \Phi_{\gamma})Y = v_1 + (\Phi_{\tilde{\gamma}} - \Phi_{\gamma'})\tilde{w}$ and to $(\Phi_{\tilde{\gamma}} - \Phi_{\gamma'})Y = v_2 + (\Phi_{\tilde{\gamma}} - \Phi_{\gamma'})\tilde{w}$,

$$(91) \quad \|(\Phi_{\tilde{\gamma}} - \Phi_{\gamma})Y\|_2 \geq A - \frac{A}{8} = \frac{7A}{8} > 0, \quad \|(\Phi_{\tilde{\gamma}} - \Phi_{\gamma'})Y\|_2 \leq \frac{A}{2} + \frac{A}{8} = \frac{5A}{8},$$

and squaring, which is legitimate because both bounds are non-negative,

$$(92) \quad \|Y\|_2^2(R_{\gamma'}^2 - R_{\gamma}^2) \geq \frac{49}{64}A^2 - \frac{25}{64}A^2 = \frac{24}{64}A^2 \geq \frac{A^2}{4} > 0.$$

Inequality (81) holds verbatim, with $\|(\Phi_{\gamma'} - \Phi_{\gamma})Y\|_2^2$ replaced by the left-hand side of (92), and repeating Steps 2 and 3 of Section A.5 with $A^2/2$ replaced by $A^2/4$ in the numerator of (85) halves the right-hand side of (86):

$$\frac{n}{2} \cdot \frac{\|Y\|_2^2(R_{\gamma'}^2 - R_{\gamma}^2)}{\|Y\|_2^2(g^{-1} + 1 - R_{\gamma}^2)} \geq \frac{n}{8} \cdot \frac{A^2}{2s^*A^2/\nu + 8n\sigma_0^2} \geq \frac{1}{2} \min\left\{\frac{n\nu}{16s^*}, \frac{n\nu^2C_{\beta}^2}{64\sigma_0^2}\right\}.$$

Substituting into (87) and using $\lambda \geq 0$ gives the third bound (27). $\square$

A.7. Proofs for Section 3.

PROOF OF LEMMA 3.2. Write $q(\gamma, \gamma')$ for the probability that $\tilde{\mathbf{P}}$ proposes γ' from γ . On $\mathcal{N}_1$ it equals $1/(2p)$, symmetric in (γ, γ') . On $\mathcal{N}_2$ the move preserves the size, so $|\gamma'| = |\gamma|$ and $q(\gamma, \gamma') = \{2|\gamma|(p - |\gamma|)\}^{-1} = q(\gamma', \gamma)$; in particular an exchange never leaves $\mathcal{M}(s_0)$. Hence q is symmetric, and for $\gamma \neq \gamma'$

$$(93) \quad \mu(\gamma) \tilde{\mathbf{P}}(\gamma, \gamma') = q(\gamma, \gamma') \min\{\mu(\gamma), \mu(\gamma')\},$$

which is symmetric; this is detailed balance for $\tilde{\mathbf{P}}$, hence for $\mathbf{P}$, and it shows $\mathbf{Q}$ is symmetric.

For irreducibility, recall that μ has full support on $\mathcal{M}(s_0)$. Given $\gamma \neq \gamma'$, delete the elements of γ one at a time and then add those of γ' one at a time; every intermediate state is a subset of γ or of γ' , hence lies in $\mathcal{M}(s_0)$, and consecutive states differ by one flip, so each transition has positive probability by (13). Aperiodicity is $\mathbf{P}(\gamma, \gamma) \geq 1/2 > 0$.

For non-negative definiteness, $\tilde{\mathbf{P}}$ is a reversible Markov kernel, hence self-adjoint on $\mathcal{L}^2(\mu)$ with spectrum in $[-1, 1]$, so $\mathbf{P} = \frac{1}{2}(\mathbf{I} + \tilde{\mathbf{P}})$ has spectrum in $[0, 1]$ and its absolute spectral gap coincides with $1 - \lambda_2$.

For (15), combine (14) and (93): $\mathbf{Q}(\gamma, \gamma') = \frac{1}{2}q(\gamma, \gamma') \min\{\mu(\gamma), \mu(\gamma')\}$, and $q(\gamma, \gamma') \geq 1/(2ps_0)$ in both cases: on $\mathcal{N}_1$ because $1/(2p) \geq 1/(2ps_0)$, and on $\mathcal{N}_2$ because $|\gamma| \leq s_0$ and $p - |\gamma| \leq p$. $\square$

A.8. Proofs for Section 4.

PROOF OF COROLLARY 4.1. For an underfitted unsaturated γ , the bound (26) is at most $-\eta \log p$ precisely when $\lambda \{\min\{\frac{n\nu}{16s^*}, \frac{n\nu^2C_{\beta}^2}{64\sigma_0^2}\} - \frac{1}{2} \log(1+g)\} \geq (\kappa + \eta) \log p$, that is when λ is at least the first term of (28), the coefficient $\min\{\frac{n\nu}{16s^*}, \frac{n\nu^2C_{\beta}^2}{64\sigma_0^2}\} - \frac{1}{2} \log(1+g)$ being positive by hypothesis. For an underfitted saturated γ , the bound (27) is at most $-\eta \log p$ precisely when $\lambda \min\{\frac{n\nu}{16s^*}, \frac{n\nu^2C_{\beta}^2}{64\sigma_0^2}\} \geq 2\eta \log p$, that is when λ is at least the second term of (28).

Taking the maximum gives (29). The overfitted claim restates (25), and (30) is obtained by evaluating the affine function at $\lambda = 0$ and $\lambda = 1$ and bounding $\frac{1}{2} \log(1 + g)$ below by $\alpha \log p$. $\square$

PROOF OF COROLLARY 4.2. Write $a := (s^* + 1)p^{1-\eta}$ and $b := p^{1-\eta}$, so that $b \leq a \leq \frac{1}{2}$ by (31).

Step 0: the drop along a path. If $\mathcal{G}^k(\bar{\gamma}) = \gamma$ then, telescoping and applying Corollary 4.1 at each of the k source states — each of which is either overfitted or underfitted, and both cases are covered by its hypotheses —

$$(94) \quad \frac{\pi_n^{(\lambda)}(\bar{\gamma} | Y)}{\pi_n^{(\lambda)}(\gamma | Y)} = \prod_{i=0}^{k-1} \frac{\pi_n^{(\lambda)}(\mathcal{G}^i(\bar{\gamma}) | Y)}{\pi_n^{(\lambda)}(\mathcal{G}^{i+1}(\bar{\gamma}) | Y)} \leq p^{-\eta k}.$$

Step 1: γ underfitted. By Lemma A.6a every $\bar{\gamma} \in \Lambda(\gamma)$ is underfitted, so Lemma A.6c gives $|\Lambda_k(\gamma)| \leq \{(s^* + 1)p\}^k$, and with (94)

$$(95) \quad \frac{\pi_n^{(\lambda)}[\Lambda(\gamma) | Y]}{\pi_n^{(\lambda)}(\gamma | Y)} \leq \sum_{k \geq 0} \{(s^* + 1)p\}^k p^{-\eta k} = \sum_{k \geq 0} a^k = \frac{1}{1-a},$$

the series converging because $a \leq \frac{1}{2}$.

Step 2: $\gamma \supseteq \gamma^$.* Split $\Lambda(\gamma) = \mathbb{M}_1 \sqcup \mathbb{M}_2$ into the precedents that contain γ^* and those that do not, so that $\gamma \in \mathbb{M}_1$ and every member of $\mathbb{M}_2$ is underfitted. By Lemma A.6c and (94),

$$(96) \quad \frac{\pi_n^{(\lambda)}[\mathbb{M}_1 | Y]}{\pi_n^{(\lambda)}(\gamma | Y)} \leq \sum_{k \geq 0} p^k p^{-\eta k} = \frac{1}{1-b}.$$

For $\bar{\gamma} \in \mathbb{M}_2$ let $f(\bar{\gamma})$ be the first state on the path $\bar{\gamma} \rightarrow \mathcal{G}(\bar{\gamma}) \rightarrow \dots \rightarrow \gamma$ that contains γ^* ; it exists because γ itself does, and it lies in $\mathbb{M}_1$ because it is a precedent of γ . Every state strictly before $f(\bar{\gamma})$ on that path is underfitted, by the definition of f , so Lemma A.6c applies to the chain from $\bar{\gamma}$ to $f(\bar{\gamma})$, whose length is at least 1 because $\bar{\gamma} \notin \mathbb{M}_1$. Grouping the sum over $\mathbb{M}_2$ by the value of f , and using (94) for the inner ratios,

$$(97) \quad \begin{aligned} \frac{\pi_n^{(\lambda)}[\mathbb{M}_2 | Y]}{\pi_n^{(\lambda)}(\gamma | Y)} &= \sum_{\bar{\gamma} \in \mathbb{M}_1} \frac{\pi_n^{(\lambda)}(\bar{\gamma} | Y)}{\pi_n^{(\lambda)}(\gamma | Y)} \sum_{\bar{\gamma}: f(\bar{\gamma})=\bar{\gamma}} \frac{\pi_n^{(\lambda)}(\bar{\gamma} | Y)}{\pi_n^{(\lambda)}(\bar{\gamma} | Y)} \\ &\leq \frac{\pi_n^{(\lambda)}[\mathbb{M}_1 | Y]}{\pi_n^{(\lambda)}(\gamma | Y)} \cdot \sum_{k \geq 1} a^k \leq \frac{1}{1-b} \cdot \frac{a}{1-a}. \end{aligned}$$

Adding (96) and (97),

$$\frac{\pi_n^{(\lambda)}[\Lambda(\gamma) | Y]}{\pi_n^{(\lambda)}(\gamma | Y)} \leq \frac{1}{1-b} \left(1 + \frac{a}{1-a}\right) = \frac{1}{(1-a)(1-b)},$$

which is the middle expression in (32). It dominates the bound $1/(1-a)$ of Step 1, so both cases are covered by it, and every $\gamma \in \mathcal{M}(s_0)$ falls into one of them: a state other than γ^* is either underfitted or overfitted, and $\gamma^* \supseteq \gamma^*$. Finally $a, b \leq \frac{1}{2}$ gives $\{(1-a)(1-b)\}^{-1} \leq 4$. $\square$

PROOF OF PROPOSITION 4.1. Identity (33) is (8) with $j = j_\gamma$, read in the direction $\log\{\pi_n^{(\lambda)}(\gamma | Y)/\pi_n^{(\lambda)}(\gamma' | Y)\} = -\log\{\pi_n^{(\lambda)}(\gamma' | Y)/\pi_n^{(\lambda)}(\gamma | Y)\}$. The right-hand side of (33) is non-positive if and only if $\lambda \log\{\mathcal{L}_n(Y | \mathcal{G}(\gamma))/\mathcal{L}_n(Y | \gamma)\} \geq \kappa \log p$, which gives

the stated equivalence; at $\lambda = 0$ it equals $\kappa \log p > 0$; and if the likelihood ratio is at most one then its logarithm is non-positive, so for $\lambda \geq 0$ the right-hand side is at least $\kappa \log p > 0$. The characterisation of λ_{crit} follows by taking the maximum over the finitely many underfitted unsaturated γ , with the convention that a state whose likelihood ratio is at most one contributes $+\infty$.

For (35), form the ratio of two copies of (5) with $\gamma' = \gamma \cup \{j_\gamma\}$:

$$\begin{aligned} \log \frac{\mathcal{L}_n(Y | \gamma')}{\mathcal{L}_n(Y | \gamma)} &= -\frac{1}{2} \log(1+g) - \frac{n}{2} \log \frac{g^{-1} + 1 - R_{\gamma'}^2}{g^{-1} + 1 - R_\gamma^2} \\ &\geq -\frac{1}{2} \log(1+g) + \min \left\{ \frac{n\nu}{16s^*}, \frac{n\nu^2 C_\beta^2}{64\sigma_0^2} \right\}, \end{aligned}$$

the last step by (81) and (86). The right-hand side being positive by hypothesis, the quotient defining λ_{crit} is at most $\kappa \log p / \{ \min \{ \frac{n\nu}{16s^*}, \frac{n\nu^2 C_\beta^2}{64\sigma_0^2} \} - \frac{1}{2} \log(1+g) \}$ for every such γ , and taking the maximum gives the claim. $\square$

A.9. Proofs for Section 5.

PROOF OF LEMMA 5.1. By Corollary 4.1 the drop bound gives $\pi_n^{(\lambda)}(\bar{\gamma} | Y) \leq \pi_n^{(\lambda)}(\mathcal{G}(\bar{\gamma}) | Y)$ at every $\bar{\gamma} \neq \gamma^*$ — each such state being either overfitted or underfitted, and both cases being covered by its hypotheses — so the inner maximum in (19) equals one, and Corollary 4.2 bounds the remaining factor by 4. Hence $\rho(\mathcal{T}) \leq 4ps_0 \cdot 4 = 16ps_0$. With $\ell(\mathcal{T}) \leq 4s_0$ from Lemma A.7b, Sinclair's bound (17) gives $\text{Gap}(\mathbf{P}_\lambda) \geq (16ps_0 \cdot 4s_0)^{-1}$, which is (36); it is the absolute spectral gap by Lemma 3.2. $\square$

PROOF OF PROPOSITION 5.1. *Step 1: the precedents of γ^* are everything.* By Lemma A.7a, every $\bar{\gamma} \in \mathcal{M}(s_0)$ satisfies $\mathcal{G}^k(\bar{\gamma}) = \gamma^*$ for some $k \geq 0$, which is the defining condition (18) for membership of $\Lambda(\gamma^*)$. Hence $\Lambda(\gamma^*) \supseteq \mathcal{M}(s_0)$, and the reverse inclusion holds by (18), so $\Lambda(\gamma^*) = \mathcal{M}(s_0)$. Since $\pi_n^{(\lambda)}(\cdot | Y)$ is a probability measure on $\mathcal{M}(s_0)$ by (6),

$$(98) \quad \pi_n^{(\lambda)}[\Lambda(\gamma^*) | Y] = \pi_n^{(\lambda)}[\mathcal{M}(s_0) | Y] = 1.$$

Step 2: Corollary 4.2 applies at $\gamma = \gamma^$.* That corollary is stated for every $\gamma \in \mathcal{M}(s_0)$, and γ^* is covered by Step 2 of its proof, which treats every γ with $\gamma \supseteq \gamma^*$; the case $\gamma = \gamma^*$ is admissible there because $\gamma^* \supseteq \gamma^*$. Substituting (98) into (32) therefore gives

$$(99) \quad \frac{1}{\pi_n^{(\lambda)}(\gamma^* | Y)} = \frac{\pi_n^{(\lambda)}[\Lambda(\gamma^*) | Y]}{\pi_n^{(\lambda)}(\gamma^* | Y)} \leq \frac{1}{\{1 - (s^* + 1)p^{1-\eta}\} \{1 - p^{1-\eta}\}}.$$

Step 3: inverting. Put $u := \{1 - (s^* + 1)p^{1-\eta}\} \{1 - p^{1-\eta}\}$, the right-hand side of (37). By (31) we have $p^{1-\eta} \leq (s^* + 1)p^{1-\eta} \leq \frac{1}{2}$, so both factors of u lie in $[\frac{1}{2}, 1]$ and hence

$$(100) \quad \frac{1}{4} \leq u \leq 1.$$

Also $\pi_n^{(\lambda)}(\gamma^* | Y) \in (0, 1]$ by (6). Inequality (99) therefore reads

$$\frac{1}{\pi_n^{(\lambda)}(\gamma^* | Y)} \leq \frac{1}{u},$$

an inequality between two elements of $(0, \infty)$. The map $x \mapsto 1/x$ is strictly decreasing on $(0, \infty)$, so applying it to both sides reverses the inequality and gives $\pi_n^{(\lambda)}(\gamma^* | Y) \geq u$, which is the first inequality of (37). The second is the left-hand inequality of (100). $\square$

PROOF OF LEMMA 5.2. Apply (7) with $\gamma' = \gamma^*$:

$$\log \frac{\pi_n^{(\lambda)}(\gamma | Y)}{\pi_n^{(\lambda)}(\gamma^* | Y)} = (|\gamma^*| - |\gamma|) \left\{ \kappa \log p + \frac{\lambda}{2} \log(1+g) \right\} + \frac{\lambda n}{2} \log \frac{g^{-1} + 1 - R_{\gamma^*}^2}{g^{-1} + 1 - R_{\gamma}^2}.$$

For the first term, $|\gamma^*| - |\gamma| \geq -|\gamma| \geq -s_0$ and the brace is non-negative, so it is at least $-s_0 \{ \kappa \log p + \frac{\lambda}{2} \log(1+g) \}$. For the second, $R_{\gamma^*}^2 \leq 1$ and $R_{\gamma}^2 \geq 0$ give

$$\frac{g^{-1} + 1 - R_{\gamma^*}^2}{g^{-1} + 1 - R_{\gamma}^2} \geq \frac{g^{-1}}{g^{-1} + 1} = \frac{1}{1+g},$$

so it is at least $-\frac{\lambda n}{2} \log(1+g)$. Adding the two and using $\lambda \geq 0$ gives (38). For (39), combine it with $\pi_n^{(\lambda)}(\gamma^* | Y) \geq \frac{1}{4}$ from Proposition 5.1. $\square$

PROOF OF THEOREM 5.1. Lemma 3.2 makes $\mathbf{P}_\lambda$ lazy and reversible, so (24) applies with $\mu = \pi_n^{(\lambda)}(\cdot | Y)$. Substitute the gap bound of Lemma 5.1 and the mass bound (39), whose logarithm is $\kappa s_0 \log p + \frac{\lambda(n+s_0)}{2} \log(1+g) + \log 4$. $\square$

A.10. Proofs for Section 6.

PROOF OF LEMMA 6.1. **a** Rules (H1), (H3) and (H4) delete a coordinate, so they decrease $|\gamma|$ and stay in $\mathcal{M}(s_0)$. Rule (H2) fires only when $\gamma \subseteq \gamma^*$, so the result is a subset of γ^* and has size at most $s^* < s_0$.

b Set

$$(101) \quad V(\gamma) := |\gamma \setminus \gamma^*| + |c_\lambda^- \setminus \gamma| + |(\gamma \cap \gamma^*) \setminus c_\lambda^+| + |\gamma \cap C_\lambda|.$$

Rule (H1) decreases the first term by one and leaves the others unchanged, since it deletes an index of S^c while $c_\lambda^-, c_\lambda^+, C_\lambda \subseteq \gamma^*$. Rule (H2) decreases the second by one and leaves the others unchanged, since $c_\lambda^- \cap C_\lambda = \emptyset$ and $c_\lambda^- \subseteq c_\lambda^+$. Rule (H3) decreases the third by one and leaves the others unchanged, the deleted index lying outside $c_\lambda^+ \supseteq C_\lambda \cup c_\lambda^-$. Rule (H4) decreases the fourth by one and leaves the others unchanged, the deleted index lying in $C_\lambda \subseteq c_\lambda^+ \setminus c_\lambda^-$. So V drops by exactly one at each step, and $V(\gamma) = 0$ forces $\gamma \subseteq \gamma^*$, $c_\lambda^- \subseteq \gamma$, $\gamma \subseteq c_\lambda^+$ and $\gamma \cap C_\lambda = \emptyset$, that is $\gamma = c_\lambda^-$; conversely $V(c_\lambda^-) = 0$ and rule (H5) fixes it. Finally $V(\gamma) \leq |\gamma| + |c_\lambda^-| \leq s_0 + s^* \leq 2s_0$, because the first, third and fourth terms of (101) count disjoint subsets of γ .

c Rules (H2), (H3) and (H4) fire only when $\gamma \subseteq \gamma^*$ and they keep the state inside γ^* , so (H1) cannot fire afterwards. Rule (H4) deletes an index of C_λ , which changes neither $c_\lambda^- \subseteq \gamma$ nor $\gamma \subseteq c_\lambda^+$, so once those two conditions hold they continue to hold. $\square$

PROOF OF THEOREM 6.1. All three are (42).

Rule (H1). Here $j = \max(\gamma \setminus \gamma^*) \in S^c$ and $\mathcal{G}_\lambda(\gamma) = \gamma \setminus \{j\}$, so (42) applied at the state $\gamma \setminus \{j\}$, whose size is $< s_0$, gives

$$\log \frac{\pi_n^{(\lambda)}(\gamma | Y)}{\pi_n^{(\lambda)}(\gamma \setminus \{j\} | Y)} = \lambda \psi_j(\gamma \setminus \{j\}) - \kappa \log p \leq \lambda(a_0 - \alpha) \log p - \kappa \log p = -\kappa_\lambda \log p$$

by Assumption 6.1 and $\lambda \geq 0$; and $\kappa_\lambda \geq \eta$ by (43).

Rules (H2) and (H3). Here $\gamma \subseteq \gamma^*$, and the state at which (42) is applied — γ under (H2), $\gamma \setminus \{j\}$ under (H3) — is contained in $\gamma^* \setminus \{j\}$. Hence (45) applies, with $j \in c_\lambda^-$ in the first case and $j \notin c_\lambda^+$ in the second, and in both $\log\{\pi_n^{(\lambda)}(\gamma | Y)/\pi_n^{(\lambda)}(\mathcal{G}_\lambda(\gamma) | Y)\} \leq -\log(2s^*)$.

Rule (H4). Identical to rule (H3), with $j \in C_\lambda$ and without the conclusion $\lambda \psi_j^+ \leq \kappa \log p - \log(2s^*)$, which is precisely what fails for an undecided covariate; the bound $\exp\{\lambda \psi_j^+ - \kappa \log p\} \leq \Theta_\lambda$ is all that remains. $\square$

PROOF OF LEMMA 6.2. Under (H1), $\bar{\gamma} = \bar{\eta} \cup \{\ell\}$ with $\ell \in S^c$: at most p choices. Under (H2), $\bar{\gamma} = \bar{\eta} \setminus \{j\}$ with $j \in c_\lambda^- \subseteq \gamma^*$: at most s^* . Under (H3), $\bar{\gamma} = \bar{\eta} \cup \{j\}$ with $j \in \gamma^* \setminus c_\lambda^+$: at most s^* . Under (H4), $\bar{\gamma} = \bar{\eta} \cup \{j\}$ with $j \in C_\lambda$: at most k_λ . $\square$

PROOF OF COROLLARY 6.1. By Lemma 6.1c the orbit of any $\bar{\gamma}$ consists of a block of (H1) steps, then a block of (H2)/(H3) steps, then a block of (H4) steps. Reading this backwards from γ , a precedent $\bar{\gamma} \in \Lambda_\lambda(\gamma)$ is obtained from γ by undoing m steps of type (H4), then k_2 of type (H2)/(H3), then k_1 of type (H1), for a unique triple (m, k_2, k_1) , and by Lemma 6.1b the deletions of (H4) and the flips of (H2), (H3) exhaust C_λ and $\gamma^* \triangle c_\lambda^-$ respectively, so $m \leq k_\lambda$ and $k_2 \leq 2s^*$.

Telescoping as in (94) and applying Theorem 6.1 at each source state, the ratio $\pi_n^{(\lambda)}(\bar{\gamma} | Y) / \pi_n^{(\lambda)}(\gamma | Y)$ is at most $\Theta_\lambda^m (2s^*)^{-k_2} p^{-\eta k_1}$, while by Lemma 6.2 the number of such $\bar{\gamma}$ is at most $k_\lambda^m (2s^*)^{k_2} p^{k_1}$. Summing over the three indices independently,

$$\begin{aligned} \frac{\pi_n^{(\lambda)}[\Lambda_\lambda(\gamma) | Y]}{\pi_n^{(\lambda)}(\gamma | Y)} &\leq \left(\sum_{m=0}^{k_\lambda} (k_\lambda \Theta_\lambda)^m \right) \left(\sum_{k_2=0}^{2s^*} 1 \right) \left(\sum_{k_1 \geq 0} p^{(1-\eta)k_1} \right) \\ &\leq (1 + k_\lambda \Theta_\lambda)^{k_\lambda} (2s^* + 1) \frac{1}{1 - p^{1-\eta}}, \end{aligned}$$

where the first sum was bounded by the binomial theorem, the second counts $k_2 \leq 2s^*$ terms each equal to one, and the third converges because $p^{1-\eta} \leq \frac{1}{2}$, which also bounds it by 2. $\square$

PROOF OF PROPOSITION 6.1. Fix γ and $j \in S^c \setminus \gamma$ and write $\Pi = \Phi_{\gamma \cup \{j\}} - \Phi_\gamma$. By (5), $\psi_j(\gamma) = -\frac{1}{2} \log(1 + g) + \frac{n}{2} \log\{(g^{-1} + 1 - R_\gamma^2)/(g^{-1} + 1 - R_{\gamma \cup \{j\}}^2)\}$, and the computation (75)–(76), which used only that Π is a rank-one projection and Lemma A.1, gives

$$\psi_j(\gamma) \leq -\alpha \log p + \frac{4\|\Pi Y\|_2^2}{\sigma_0^2},$$

using $\frac{1}{2} \log(1 + g) \geq \alpha \log p$ from Assumption 3.3. Now $\Pi Y = \Pi X_{\gamma^*} \beta_{\gamma^*}^* + \Pi \tilde{w}$ by (57), so by (58), Lemma A.5 applied to $\gamma \cup \{j\}$ and j , and Lemma A.2b applied to the nested pair $\gamma \subset \gamma \cup \{j\}$,

$$\|\Pi Y\|_2^2 \leq 2\|\Pi X_{\gamma^*} \beta_{\gamma^*}^*\|_2^2 + 2\|\Pi \tilde{w}\|_2^2 \leq 2n\omega(X)\|\beta_{\gamma^* \setminus \gamma}^*\|_2^2 + 4(L + \tilde{L})\sigma_0^2 \log p,$$

and $\|\beta_{\gamma^* \setminus \gamma}^*\|_2^2 \leq \|\beta_{\gamma^*}^*\|_2^2$. Multiplying by $4/\sigma_0^2$ gives (49). $\square$

A.11. Proofs for Section 7.

PROOF OF LEMMA 7.1. a None of (H1)–(H3) changes $\gamma \cap C_\lambda$, by the three sentences that prove Lemma 6.1b, so F_λ preserves B_J . The function (101) without its fourth term drops by exactly one at each step of F_λ and vanishes precisely when $\gamma \subseteq \gamma^*$, $c_\lambda^- \subseteq \gamma$ and $\gamma \subseteq c_\lambda^+$; within B_J the only such state is $c_\lambda^- \cup J = v_J$. Hence every $\gamma \in B_J$ reaches v_J , which is a.

b The three drop bounds of Theorem 6.1 for rules (H1)–(H3) are all at most $\max\{p^{-\eta}, (2s^*)^{-1}\}$, which is at most 1; the length and single-flip claims are Lemma 6.1a and $V \leq s_0 + s^* \leq 2s_0$.

c Run the proof of Corollary 6.1 for F_λ at the state v_J . Since rule (H4) is absent, the third factor is missing and the first two give $\pi_n^{(\lambda)}[\Lambda^{F_\lambda}(v_J) | Y] \leq 2(2s^* + 1) \pi_n^{(\lambda)}(v_J | Y)$. By a the left-hand side is $\bar{\pi}_\lambda(J)$. $\square$

PROOF OF THEOREM 7.1. Every edge of $\tilde{\mathcal{T}}_\lambda$ is a single flip, so Lemma 3.2 gives $\mathbf{Q}_\lambda(e) \geq \frac{1}{4p} \min\{\pi_n^{(\lambda)}(\bar{\gamma} | Y), \pi_n^{(\lambda)}(\bar{\gamma}' | Y)\}$ for its two endpoints. There are two kinds.

Edges of segments (i) and (iii). Such an edge is $e = (\bar{\gamma}, F_\lambda(\bar{\gamma}))$ for some $\bar{\gamma}$, and by Lemma 7.1b it is uphill, so the minimum is $\pi_n^{(\lambda)}(\bar{\gamma} | Y)$. A pair (γ, γ') uses e in the direction of F_λ only within its segment (i), where the argument of Lemma A.7c applies unchanged and forces $\gamma \in \Lambda^{F_\lambda}(\bar{\gamma})$; the opposite orientation is the mirror image and gives the same bound because $\mathbf{Q}_\lambda$ is symmetric. Hence, using $\sum_{\gamma'} \pi_n^{(\lambda)}(\gamma' | Y) = 1$ and Lemma 7.1c applied at $\bar{\gamma}$ in place of v_J , that is the bound $\pi_n^{(\lambda)}[\Lambda^{F_\lambda}(\bar{\gamma}) | Y] \leq 2(2s^* + 1)\pi_n^{(\lambda)}(\bar{\gamma} | Y)$ established in its proof,

$$\frac{1}{\mathbf{Q}_\lambda(e)} \sum_{\tilde{T}_{\gamma, \gamma'} \ni e} \pi_n^{(\lambda)}(\gamma | Y) \pi_n^{(\lambda)}(\gamma' | Y) \leq 4p \frac{\pi_n^{(\lambda)}[\Lambda^{F_\lambda}(\bar{\gamma}) | Y]}{\pi_n^{(\lambda)}(\bar{\gamma} | Y)} \leq 8p(2s^* + 1).$$

Edges of segment (ii). Such an edge is $e = (v_J, v_{J \triangle \{j\}})$ with $j \in C_\lambda$, and it is used by $(\gamma, \gamma') \in B_{J_1} \times B_{J_2}$ exactly when the increasing-order path from J_1 to J_2 traverses $(J, J \triangle \{j\})$. Summing $\pi_n^{(\lambda)}(\gamma | Y) \pi_n^{(\lambda)}(\gamma' | Y)$ over each block gives $\bar{\pi}_\lambda(J_1) \bar{\pi}_\lambda(J_2)$, so the numerator is the inner sum of (51). For the denominator, Lemma 7.1c gives

$$\mathbf{Q}_\lambda(e) \geq \frac{1}{4p} \min\{\pi_n^{(\lambda)}(v_J | Y), \pi_n^{(\lambda)}(v_{J \triangle \{j\}} | Y)\} \geq \frac{\min\{\bar{\pi}_\lambda(J), \bar{\pi}_\lambda(J \triangle \{j\})\}}{8p(2s^* + 1)},$$

so the congestion at e is at most $8p(2s^* + 1)\varrho_\lambda$.

Length. Segments (i) and (iii) have at most $2s_0$ edges each by Lemma 7.1b and segment (ii) at most $k_\lambda \leq s^* \leq s_0$, so $\ell(\tilde{\mathcal{T}}_\lambda) \leq 5s_0$. Sinclair's bound and Lemma 3.2 now give (52). $\square$

PROOF OF LEMMA 7.2. Write $\bar{\pi}_\lambda = \bigotimes_{i=1}^{k_\lambda} \mu_i$ and fix the directed edge $(J, J \triangle \{j\})$. A pair (J_1, J_2) traverses it exactly when, at that moment, the coordinates below j have been flipped and those above have not: that is, J_2 agrees with J below j and takes the value $1 - J_j$ at j , while J_1 agrees with J at j and above. So J_1 is free below j and J_2 is free above j , and by independence

$$\begin{aligned} \sum_{(J_1, J_2)} \bar{\pi}_\lambda(J_1) \bar{\pi}_\lambda(J_2) &= \left[\sum_a \prod_{i < j} \mu_i(a_i) \right] \prod_{i \geq j} \mu_i(J_i) \\ &\quad \cdot \prod_{i < j} \mu_i(J_i) \mu_j(1 - J_j) \left[\sum_b \prod_{i > j} \mu_i(b_i) \right], \end{aligned}$$

the two bracketed sums being 1. The remaining product is $\bar{\pi}_\lambda(J) \mu_j(1 - J_j)$. Since $\bar{\pi}_\lambda(J \triangle \{j\}) = \bar{\pi}_\lambda(J) \mu_j(1 - J_j) / \mu_j(J_j)$, the quotient in (51) equals

$$\frac{\bar{\pi}_\lambda(J) \mu_j(1 - J_j)}{\bar{\pi}_\lambda(J) \min\{1, \mu_j(1 - J_j) / \mu_j(J_j)\}} = \max\{\mu_j(J_j), \mu_j(1 - J_j)\} \leq 1. \quad \square$$

PROOF OF LEMMA 7.3. The inner sum of (51) is over the same index set for $\bar{\pi}_\lambda$ and $\bar{\mu}$, and each of its terms is a product of two masses, so it is at most e^{2D} times the corresponding sum for $\bar{\mu}$; the denominator is at least e^{-D} times the corresponding minimum. Apply Lemma 7.2 to $\bar{\mu}$. $\square$

PROOF OF PROPOSITION 7.1. Let $\bar{\mu}$ be the product measure on 2^{C_λ} with odds $\theta_j := \exp\{\lambda \psi_j(c_\lambda^-) - \kappa \log p\}$, normalised, so that $\bar{\mu}(J) \propto \prod_{j \in J} \theta_j$. Enumerating $J = \{j_1 < \dots <$

$j_m\}$ and telescoping (42) along $c_\lambda^- \subset c_\lambda^- \cup \{j_1\} \subset \dots \subset v_J$,

$$\log \frac{\pi_n^{(\lambda)}(v_J | Y)}{\pi_n^{(\lambda)}(v_\emptyset | Y)} = \sum_{i=1}^m \{ \lambda \psi_{j_i}(c_\lambda^- \cup \{j_1, \dots, j_{i-1}\}) - \kappa \log p \},$$

and each summand differs from $\log \theta_{j_i}$ by at most $\lambda \Delta_\lambda^C \leq \Delta_\lambda^C$ by (54), the contexts involved being of the form $c_\lambda^- \cup U$ with $U \subseteq C_\lambda \setminus \{j_i\}$. Writing $\tilde{\mu}(J) := \prod_{j \in J} \theta_j$ and $\tilde{g}(J) := \pi_n^{(\lambda)}(v_J | Y) / \pi_n^{(\lambda)}(v_\emptyset | Y)$ for the two unnormalised weights, the display therefore gives

$$(102) \quad \left| \log \frac{\tilde{g}(J)}{\tilde{\mu}(J)} \right| \leq |J| \Delta_\lambda^C \leq k_\lambda \Delta_\lambda^C =: D_1 \quad \text{for every } J \subseteq C_\lambda.$$

Summing (102) over J , the two normalising constants $\sum_J \tilde{g}$ and $\sum_J \tilde{\mu} = Z_\mu$ also differ by a factor at most e^{D_1} , so the two *normalised* measures differ by a factor at most e^{2D_1} pointwise.

Next, Lemma 7.1c and $\pi_n^{(\lambda)}(v_J | Y) \leq \bar{\pi}_\lambda(J)$ give $\bar{\pi}_\lambda(J) = \pi_n^{(\lambda)}(v_J | Y) c(J)$ with $c(J) \in [1, 2(2s^* + 1)]$; both $\bar{\pi}_\lambda$ and the normalisation of $\tilde{g}$ being probability measures, they differ pointwise by a factor at most $\{2(2s^* + 1)\}^2$, one factor from $c(J)$ and one from the normaliser.

Combining the two paragraphs, Lemma 7.3 applies with $D = 2k_\lambda \Delta_\lambda^C + 2 \log\{2(2s^* + 1)\}$, and $\varrho_\lambda \leq e^{3D}$ is exactly (55). $\square$